Arab Academy for Science, Technology and Maritime Transport

College of Engineering and Technology

Electrical Energy Engineering

B. Sc. Final Year Project

SWAPPABLE BATTERIES STATION

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DECLARATION

We hereby certify that this material, which we now submit for assessment on the programme of study leading to the award of Bachelor of Science in Electrical Energy Engineering is entirely our own work, that we have exercised reasonable care to ensure that the work is original, and does not to the best of our knowledge breach any law of copyright, and has not been taken from the work of others save and to the extent that such work has been cited and acknowledged within the text of our work.

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ACKNOWLEDGMENT

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ABSTRACT

This comprehensive report dives into the topic of electric scooters and batteries, highlighting the challenges and solutions associated with their usage. The report provides a detailed explanation of the concept of swappable battery stations and how they can be used to overcome the limitations of electric scooter batteries.

The report also provides insights into the various steps involved in the project, starting from the market research and identification of suitable battery types and electric scooters available in Egypt. The report presents case studies and iterative designs to showcase the development process, with a focus on the mini model and its features.

Moreover, the report delves into the mobile application and website developed to connect users with the swappable battery station, outlining its benefits and functionality in detail. The report also provides a breakdown of the components used in the project, including their purpose and cost.

In addition, the report sheds light on the challenges faced during the project, providing a critical analysis of various issues encountered and the steps taken to overcome them.

Overall, this report offers a comprehensive overview of the development of swappable battery stations for electric scooters, providing insights into the market research, design process, and challenges faced. It highlights the potential benefits of swappable battery stations, including reduced costs, increased convenience, and reduced environmental impact, making them a promising solution for the widespread adoption of electric scooters.

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LIST OF ACRONYMS/ABBREVIATIONS

ABBREVIATION	DEFINITION
A	Ampere
Ah	Ampere hour
EGP	Egyptian pound
Km/hr	Kilometer per hour
Kw	Kilo watt
Km	Kilometer
W	Watt
V	Volt
MQTT	Message Queuing Telemetry Transport
RED	Rapid Eye Developer

LIST OF UTILIZED STANDARDS

IEC 62133 "Part 2": Safety testing for lithium-ion batteries.

IEC 60335: Safety of electric appliances for household and similar purposes.

IEC 62660: Specifies performance and life testing of lithium-ion cells used for propulsion of electric vehicles.

LIST OF REALISTIC CONSTRAINTS

- **COST:** The project's high cost was a significant challenge at the outset. To address this issue, the team applied for a fund to help cover expenses. They also looked for ways to reduce costs wherever possible, such as by selecting more affordable components or using open-source software.
- **LIMITED NUMBER OF SWAPPABLE BATTERY SCOOTERS IN EGYPT:** This was a significant obstacle, as the team needed swappable batteries to make their system work. To overcome this challenge, they searched extensively for electric scooters and eventually found a few models that met their requirements. However, they still had to import some of the components from other countries.
- **LACK OF INFORMATION:** The team found it challenging to obtain detailed information about chargers and batteries in Egypt, as there was a lack of documentation and most salespeople were not knowledgeable about the products they were selling. To overcome this challenge, the team had to conduct extensive research and testing to determine which components would work best with their system.
- **CODING CHALLENGES:** The team faced several coding challenges, a significant amount of time was dedicated to writing the codes for the mobile application and website, as we were not familiar with the languages, they were written in. Due to our lack of expertise in these languages, we had to spend a considerable amount of time learning and practicing in order to write the codes effectively.
- **THEFT AND VANDALISM:** The team recognized that theft and vandalism were significant risks for their system, as the batteries could be valuable targets. To mitigate these risks, they implemented a security system that requires users to provide identification and other information to ensure their identity. They also plan to install swappable battery stations in secure locations, such as those with security cameras or personnel.

Overall, the team faced many challenges in developing their swappable battery system for electric scooters. However, they were able to overcome these challenges through creative problem-solving, extensive research and testing, and a commitment to finding solutions that met their needs.

C h a p t e r O n e

1 INTRODUCTION

1.1 OVERVIEW

The objective of this graduation project is to tackle the critical environmental issue of air pollution, which has negative impacts on both human health and the ecosystem. The project's main focus is to investigate alternative transportation options, specifically electric scooters, as a viable solution for reducing air pollution and traffic congestion. Electric scooters are a promising option since they are cost-effective, space-efficient, and emit zero emissions. Thus, they are an ideal choice for businesses such as restaurants, pharmacies, and shops seeking to contribute to environmental sustainability. The project aims to address the issues associated with electric scooters, specifically the charging process, by developing a unit that can charge swappable electric batteries, making the charging process shorter and more convenient. This way, the project aims to promote and facilitate the use of electric scooters.

1.2 PROBLEM

The main problem addressed in this project is the detrimental impact of air pollution caused by vehicle emissions, particularly in densely populated areas. The combustion of gasoline and diesel fuels in traditional vehicles releases harmful pollutants such as nitrogen oxides, particulate matter, carbon monoxide, hydrocarbons, benzene, and formaldehyde into the air. These pollutants have severe consequences for both human health and the environment.

From a human health perspective, exposure to air pollution can lead to respiratory problems, cardiovascular diseases, and an increased risk of cancer. The most vulnerable groups, including children, the elderly, and individuals with pre-existing health conditions, are at higher risk of experiencing adverse health effects. Furthermore, air pollution contributes to the development and exacerbation of respiratory conditions such as asthma and chronic obstructive pulmonary disease (COPD).

In addition to its impact on human health, air pollution also poses a significant threat to the environment. Pollutants released by vehicles contribute to the formation of smog, which reduces visibility and has detrimental effects on ecosystems. Acid rain, caused by the deposition of nitrogen oxides and sulphur dioxide, damages vegetation, aquatic ecosystems, and infrastructure. Furthermore, the emission of greenhouse gases, including carbon dioxide (CO2) from vehicle combustion, contributes to global warming and climate change, leading to rising temperatures, sea-level rise, and extreme weather events.

The transportation sector is a major contributor to air pollution, accounting for a significant portion of greenhouse gas emissions. In urban areas, traffic congestion exacerbates the problem, as vehicles are frequently idling or operating in stop-and-go conditions, resulting in increased emissions. Traditional vehicles powered by fossil fuels, such as gasoline and diesel, emit substantial amounts of pollutants, contributing to both local and global air pollution.

To address this complex problem, the project focuses on exploring electric scooters as a viable solution. Electric scooters offer numerous advantages over traditional vehicles, particularly in densely populated areas. They produce zero direct emissions, as they are powered by electricity instead of fossil fuels. By utilizing electric scooters for deliveries and short-distance transportation needs, businesses, such as restaurants, pharmacies, and shops, can significantly reduce their environmental impact.

However, several challenges must be addressed to facilitate the widespread adoption of electric scooters. These include the establishment of adequate charging infrastructure, range limitations, public perception and acceptance, regulatory frameworks, and the availability of affordable and reliable electric scooter models. Additionally, the project will consider factors such as cost-effectiveness, operational efficiency, and user experience to evaluate the feasibility and practicality of electric scooters as a sustainable transportation option.

By investigating the potential of electric scooters and addressing these challenges, the project aims to contribute to the broader goal of reducing air pollution, improving air quality, mitigating climate change, and promoting sustainable urban mobility.

1.3 PROCEDURE

The project will follow a systematic procedure to investigate the effectiveness of electric scooters in reducing air pollution and traffic congestion. The procedure includes the following steps:

- **Research and Literature Review:** Conduct an in-depth study of existing research, literature, and initiatives related to air pollution, transportation, and electric scooters. This will provide a comprehensive understanding of the current state of the problem and potential solutions.
- **Data Collection and Analysis:** Gather relevant data on air pollution levels, traffic congestion, and the environmental impact of traditional vehicles in the project's target area. Analyse the collected data to assess the extent of the problem and identify areas that can benefit from the implementation of electric scooters.
- **Case Studies and Pilot Programs:** Conduct case studies and pilot programs to evaluate the feasibility and effectiveness of electric scooters as an alternative mode of transportation. Collaborate with businesses, such as restaurants, pharmacies, and shops, to assess the practicality and impact of using electric scooters for deliveries and transportation needs.

- **Evaluation and Results:** Analyse the data and feedback collected during the pilot programs to evaluate the environmental benefits, cost-effectiveness, and overall performance of electric scooters. Compare the results with traditional vehicles to quantify the reduction in air pollution and traffic congestion achieved by adopting electric scooters.
- **Recommendations and Future Directions:** Based on the findings, provide recommendations for promoting the use of electric scooters in the project's target area. Identify potential barriers, policy implications, and strategies to encourage individuals, businesses, and governments to embrace electric scooters as a sustainable transportation solution. Outline future research directions to further enhance the implementation and effectiveness of electric scooters.

By following this procedure, the project aims to contribute to the broader goal of reducing air pollution and improving environmental sustainability through the adoption of electric scooters as an alternative mode of transportation.

C h a p t e r T w o

2 LITERATURE REVIEW

2.1 SWAPPABLE BATTERY STATIONS

The main barrier which faces the widespread adoption of electric scooters is their limited range and the time required to fully charge their batteries. Consumers are accustomed to gasoline-powered vehicles and the convenience of being able to refuel at any gas station. However, the fear of running out of charge before reaching their destination is a common concern among potential electric scooter buyers. This problem is particularly evident when the battery charge drops to zero percent.

This project aims to address these obstacles by introducing swappable battery stations. A swappable battery station is essentially a cabinet with multiple chambers, each of which is locked. The dimensions of each chamber correspond to those of an electric scooter battery. A mobile application and website is developed to help users locate the nearest battery station, check the availability and state of the batteries, and reserve a battery for a specific period.

When a user arrives at the station, they remove their discharged battery, scan the QR code on the chamber door, and place the battery inside. The user then plugs the battery into the charger and locks the chamber door. The battery's health, efficiency, serial number, temperature, and authenticity are checked before it begins to charge. If all the criteria are met, the battery will start to charge, and the next chamber containing a fully charged and reserved battery will open. The user can then unplug the battery and place it in their scooter before closing the door. The entire process should take no longer than five minutes, eliminating the issue of long charging times.

The spread of electric vehicles is on the rise globally due to the increased awareness of the environmental impact of gasoline-powered vehicles and the advancements in battery technology. Governments around the world are promoting the use of clean energy and reducing carbon emissions. Additionally, major automakers are investing heavily in the production of electric vehicles, resulting in a wider range of models with improved performance and longer ranges. As a result, the number of electric vehicles on the road has continued to increase steadily, and it is expected that the trend will continue in the coming years as more people opt for sustainable transportation options.

The trend towards the spread of electric vehicles is not limited to any particular region or country. In fact, it is a global phenomenon, with countries like Norway, Iceland, and Sweden leading the way in terms of electric vehicle adoption. In Norway, for instance, electric vehicles account for over 50% of new car sales, thanks to generous government incentives that include exemptions from taxes, tolls, and ferry fees. Similarly, Iceland has one of the highest electric vehicle ownership rates in the world, with over 25% of all cars on the road being electric.

In addition to government incentives, the growth in electric vehicle sales is also being driven by the increasing number of charging stations that are being deployed globally. This has helped to alleviate range anxiety concerns among potential electric vehicle buyers and has made it easier for electric vehicle owners to travel longer distances.

Moreover, the adoption of electric vehicles is not limited to personal vehicles. Electric buses, trucks, and commercial vehicles are also gaining popularity, especially in urban areas where air pollution is a major concern. Many cities around the world have begun to phase out diesel-powered buses in favor of electric buses, which are quieter, emit less pollution, and have lower operating costs.

Overall, the spread of electric vehicles is a positive development for the environment, as it helps to reduce greenhouse gas emissions and other pollutants that are harmful to human health. As battery technology continues to improve, electric vehicles will become even more attractive to consumers, and it is likely that we will see even more widespread adoption in the future.

While the adoption of electric vehicles is growing rapidly, there are still some challenges that need to be addressed to attain widespread adoption. The most highlighted challenge is range anxiety. Despite the increasing number of charging stations, the limited range of electric vehicles remains a major concern for many potential buyers. While the range is improving, it is still not comparable to that of gasoline-powered vehicles, and long charging times can also be a deterrent.

As previously stated, the swappable battery station addresses two significant challenges that hinder the adoption of electric vehicles. By providing ubiquitous stations for battery swapping, it effectively resolves the issues of extended charging time and range anxiety.

In Egypt, numerous startups have emerged that offer customized electric mobility solutions to enable businesses to switch to electric vehicles at a significantly lower cost than purchasing a new EV. One such company is Shift EV, which is a Cairo-based startup specializing in electric mobility technology. They produce cutting-edge lithium-ion battery packs that can be easily integrated with existing vehicle chassis. Additionally, they provide remote connectivity to fleet vehicles through their proprietary fleet operating system, Shift Ware, which offers complete digital oversight and management.

As of the searches till September 2021, there were several countries where swappable battery stations were being implemented or planned. The first country is Taiwan. Gogoro, is a Taiwanese electric scooter company that has developed a unique swappable battery system for its vehicles. The company has built a network of over 2,000 battery swapping stations in Taiwan, known as "GoStations," which are located throughout the country. Gogoro owners can swap out their depleted battery at a GoStation for a fully charged one in a matter of seconds, eliminating the need for long charging times and making it more convenient for riders to use their electric scooters for longer distances.

Figure 2-1: Gogoro swappable battery stations. [4]

In France the French company, Bolloré, is a company that operates a car-sharing service using its Bluecars, which are electric vehicles with a swappable battery system. The company has implemented a network of battery swapping stations in several French cities, including Paris, Lyon, and Bordeaux. Bluecar drivers can swap out their depleted battery for a fully charged one at a Bolloré battery swapping station, which takes only a few minutes. This allows the company to keep its fleet of vehicles running for longer periods without having to worry about charging times.

Figure 2-2: Bolloré swappable battery stations. [5]

In India, SUN Mobility is an Indian startup that has developed a swappable battery system for electric two-wheelers and three-wheelers. The company has partnered with several vehicle manufacturers and fleet operators to expand its network of battery swapping stations in India. SUN Mobility's battery swapping system is designed to be compatible with a wide range of electric vehicles, making it a more flexible solution for businesses and consumers looking to switch to electric mobility.

Figure 2-3: SUN Mobility swappable battery stations. [6]

In China, NIO, is a Chinese electric vehicle manufacturer that has developed a battery swapping system for its electric cars. NIO has implemented a network of battery swapping stations in several Chinese cities, including Beijing, Shanghai, and Shenzhen. NIO owners can swap out their depleted battery for a fully charged one at a NIO battery swapping station in a matter of minutes, making it more convenient for them to use their electric cars for longer distances.

Figure 2-4: NIO swappable battery stations. [7]

In Japan, Honda has developed a battery swapping system for its electric two-wheelers and has implemented a network of battery swapping stations in Japan. Honda's battery swapping system is called "Mobile Power Pack," and it allows users to swap out their depleted battery for a fully charged one at a Honda battery swapping station. Honda has partnered with several companies in Japan to expand its network of battery swapping stations, making it easier for users to use their electric two-wheelers for longer distances.

Figure 2-5: Honda swappable battery stations. [8]

Our project is the first of its kind to be implemented in Egypt, and it aims to promote the use of electric scooters while minimizing pollution. By introducing this innovative solution, we hope to facilitate the adoption of electric mobility in Egypt.

2.2 BATTERIES

To ensure that the project was feasible and effective, we began by conducting extensive market research to identify all the available electric scooters and compatible battery types in Egypt. After a thorough analysis, we identified three potential battery types that could fulfil our project requirements: lead acid, nickel metal hydride (NiMH), and lithium-ion batteries.

After careful consideration, lead acid batteries were eliminated due to their weight and bulky size. Additionally, lead acid batteries are made of toxic materials, and overcharging can cause an increase in pressure inside the battery, making them unsuitable for our project. Though nickel metal hydride batteries are commonly used in electric cars, we decided against using them due to their high cost, high self-discharge rate, low temperature operation, and higher cooling requirements.

Figure 2-6: Lead-acid battery.

Figure 2-7: Nickel metal hydride battery.

Lithium-ion batteries were ultimately chosen for the project due to several reasons. Lithium-ion batteries are lightweight, compact, and portable. They have a high energy density, are eco-friendly, require low maintenance, have more charge cycles, and have a low self-discharge rate. However, it is important to note that lithium-ion batteries are more expensive than other options, and complete discharge can damage the battery. Additionally, the transportation of lithium-ion batteries is limited to ships, as they are not permitted to be sent by airplanes due to safety concerns.

Figure 2-8: Lithium-ion battery. [12]

Overall, the research done allowed us to make informed decisions about the most suitable battery type for our project. By selecting lithium-ion batteries, we were able to ensure that our swappable battery station would be lightweight, portable, and eco-friendly while also providing a high energy density and low maintenance requirements, making it an optimal solution for electric scooter users. Despite the higher cost of lithium-ion batteries, their numerous benefits and advantages make them an ideal choice for our project.

It is worth noting that the selection of the appropriate battery type is critical in the development of electric vehicles and their charging infrastructure. As electric vehicles continue to gain popularity and become more wide-spread, the demand for efficient and reliable battery technology will only increase. Therefore, it is important to continue researching and developing new and improved battery technologies to meet the growing demand for electric vehicles and charging infrastructure. The following table summarizes the advantages and disadvantages of lithium-ion batteries.

Table 2-1: Advantages and disadvantages of lithium-ion batteries.

Advantages and disadvantages of lithium ion batteries	
Advantages	*Disadvantages*
Light weight & Compact	High cost
Portable	Complete discharge damage battery
High energy density	Transportation limited to ship
Eco-friendly	
Low maintenance	
More charge cycles	
Low self-discharge rate	

2.3 ELECTRIC SCOOTERS

To ensure that the swappable battery station project is a success, we conducted extensive research on the available electric scooters in Egypt. We visited showrooms and inquired about all the available options to identify the most suitable electric scooter that fulfilled all our project requirements. However, we found that many of the available options were not suitable for our needs.

The first option was the Yadea c-umi, which had a range of 50-60 km and a battery capacity of 60V/21Ah. It was priced at 27,000 LE and had a charging time of 5-6 hours. Unfortunately, all the available Yadea scooters in Egypt used lead-acid batteries, making them unsuitable for our swappable battery station project.

Figure 2-9: Yadea c-umi scooter.

The second option was the Alpha Kader City Hunter, which was priced at 18,000 LE and had an average range of 90 km, charging time of 6 hours, and a battery capacity of 72V/20Ah. However, like the Yadea c-umi, the Alpha Kader City Hunter also used lead-acid batteries, making them difficult to swap easily.

Figure 2-10: Alpha Kader City Hunter scooter.

Also, the Miku Max electric scooter was considered, which had a reasonable price of 22,000 LE, a charging time of 4 hours, and a range of 60 km. It was equipped with an 800W Bosch motor and a 60V/20Ah lithium-ion battery. However, its production had stopped in Egypt due to importation issues, making it unavailable for our project.

Figure 2-11: Miku Max scooter.

Another promising option was the Barq Rena Max, which had a charging time of 2 hours, a range of 150 km, and a battery capacity of 5.6 kWh. However, it was set to be released in 2025, which was beyond the scope of our project timeline.

Figure 2-12: Barq Rena Max scooter. [11]

Despite the numerous options available, we ultimately selected the Glide G1 electric scooter for our project. This scooter was equipped with a 60V/20Ah lithium-ion battery, had a range of 55 km, and a charging time of 4 hours. It was also equipped with an 800W Bosch motor, had a top speed of 45 km/hr, and was priced at 31,000 LE.

Figure 2-13: Glide G1 scooter. [10]

The Glide G1 electric scooter was identified as the optimal choice for our project due to its numerous benefits and advantages. Its lithium-ion battery was compatible with our swappable battery station, and it had a suitable range and charging time. Additionally, its lightweight and compact design made it easy to handle and transport.

It is worth noting that the selection of the appropriate electric scooter is a critical aspect of developing a swappable battery station project. The electric scooter must be compatible with the battery type used in the station, have a suitable range and charging time, and be user-friendly and cost-effective.

C h a p t e r Three

3 MARKET RESEARCH

The market research chapter delves into the surveys and case studies conducted to support the project's idea, assess its applicability, and evaluate its feasibility. This chapter aims to provide valuable insights into the current market landscape, consumer preferences, and the viability of implementing electric scooters as a sustainable transportation solution. Through comprehensive research and analysis of surveys and case studies, this chapter seeks to explore the potential market demand, identify key challenges, and propose strategies for successful implementation. By examining the practical aspects of the project, this research contributes to understanding the market dynamics and informs decision-making processes for the adoption of electric scooters in promoting sustainable urban mobility.

3.1 SURVEY

Electric scooters have gained increasing popularity in recent years due to their eco-friendliness, affordability, and convenience. However, despite their many benefits, the adoption of electric scooters has been somewhat slow, with many consumers still opting for traditional gasoline-powered vehicles. To better understand the factors that influence consumer behavior and attitudes towards electric scooters, we conducted a comprehensive survey that asked participants about their age, gender, driving habits, ownership of electric vehicles, and their willingness to consider purchasing an electric scooter. Additionally, we asked participants about the challenges they perceived with owning an electric scooter, such as the cost of maintenance, range anxiety, and charging infrastructure. We also asked whether participants would be more inclined to purchase an electric scooter if these challenges were addressed. Our objective was to identify the key drivers and barriers to the adoption of electric scooters and to gain insight into the market potential for this type of vehicle. By analyzing the results of the survey, we hope to gain a better understanding of consumer preferences and behaviors towards electric scooters and to inform future efforts to promote sustainable transportation. The findings of this survey could also have broader implications for the broader transportation industry, as electric scooters could potentially serve as an alternative to traditional automobiles in urban environments.

Based on the survey we conducted, the most common challenges that participants perceived with owning an electric scooter were the cost of maintenance, range anxiety, and the availability of charging infrastructure. Many participants were concerned about the potential expenses associated with repairing or replacing batteries, as well as the overall cost of ownership, which they perceived as higher than that of traditional gasoline-powered scooters. Range anxiety, or the fear of running out of battery power before reaching their destination, was also a common concern among participants. They were worried about the limited range of electric scooters and the inconvenience of having to stop and recharge frequently. Lastly, the availability of charging infrastructure was a significant concern for participants. Many felt that charging stations were not widely available, making it challenging to use electric scooters for longer commutes or trips. Overall, these challenges were perceived as significant barriers to the adoption of electric scooters, and addressing them would be crucial in promoting the use of electric scooters as a sustainable mode of transportation.

As age can significantly impact consumer behavior and attitudes towards new technology, we included a question about age in our survey on electric scooters. Our goal was to gain insight into whether age plays a significant role in the decision to purchase an electric scooter and to identify any age-related trends or patterns. By analyzing the age demographics of our survey respondents, we aimed to better understand the market potential for electric scooters and to inform future efforts to promote sustainable transportation. Understanding the attitudes and preferences of different age groups towards electric scooters is essential in developing targeted marketing and outreach strategies to promote the adoption of this emerging technology. Individuals with varying ages, ranging from 16 years old to over 60+, provided different responses.

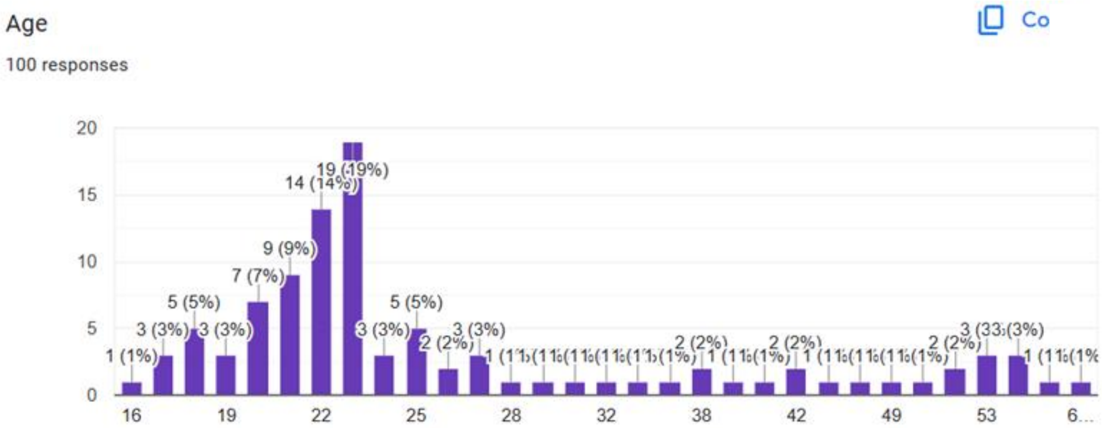

Figure 3-1: Age chart.

Gender information was gathered to gain a deeper insight into the perspectives and experiences of various groups. The data revealed that 52.8% of respondents identified as female, while 47.2% identified as male.

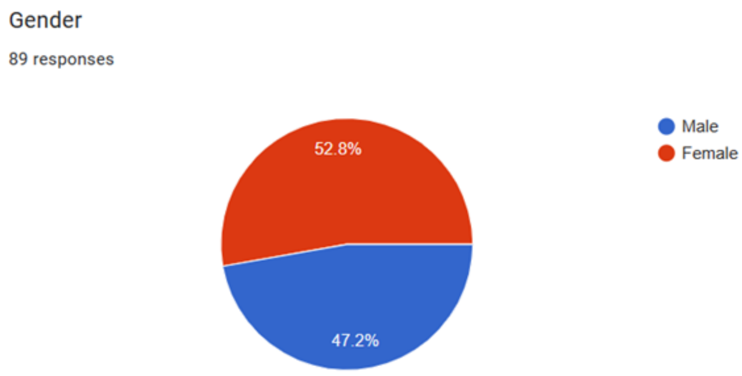

Figure 3-2: Gender chart.

As part of our survey, an interesting question was in learning about people's transportation habits, including whether or not people drive. This information will help better understand the transportation needs and preferences of respondents. The answers provided to this question was yes we drive a car, or yes we drive a scooter or motorcycle, or no. 69.7% responded with yes we drive a car. 28.1% responded with no we don't drive. The remaining 2.2% responded with yes we drive a scooter or motorcycle. This question showed how much the swappable battery station project would promote the usage of scooters and decrease the usage of cars which will lead to decreasing traffic and carbon dioxide emissions.

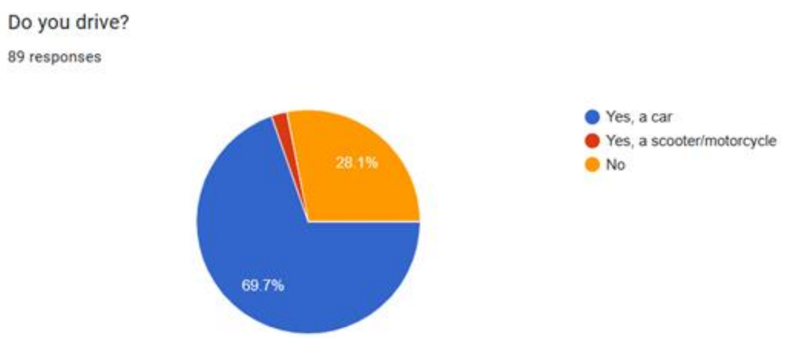

Figure 3-3: Percentage of people who drive.

Do you have an electric vehicle was a very concerning question as to gather information about whether electric vehicles are common or not. The options in the answers were electric cars, electric scooter or motorcycles, both, or none. 93% responded with none. 6% responded with electric scooters or motorcycles. 1% responded with electric car.

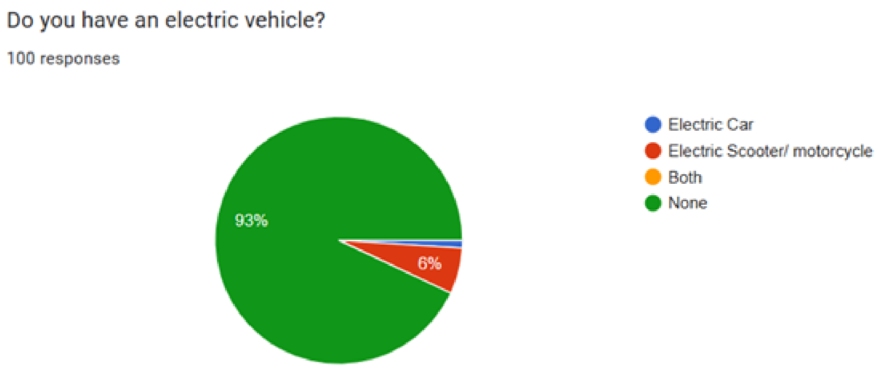

Figure 3-4: Number of electric vehicles owned.

The survey also included a crucial question about the willingness of respondents to purchase electric scooters. The results showed that 51% answered in the affirmative, while 49% answered in the negative. These findings suggest that individuals may be more inclined to purchase electric scooters once the challenges associated with their adoption are addressed.

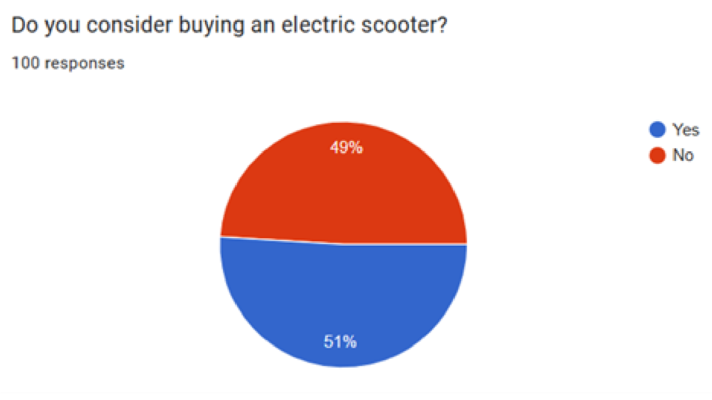

Figure 3-5: Willingness to buy electric scooters.

People answering questions in the survey were then asked why they would consider buying electric scooters. Most of the people who responded with stated the reason was that it is dangerous. Other responds were mostly that it is ecofriendly, better for the environment, saves a lot of money, interesting, and other people said that it looked fun.

If a solution was found for the challenges facing electric scooters (such as the small range and the charging time), would you buy an electric scooter? Was the next question in the survey and the answers to this question were 73% yes and the other 27% no. which meant that swappable battery station project would benefit the spread of electric scooters a lot and would benefit the environment.

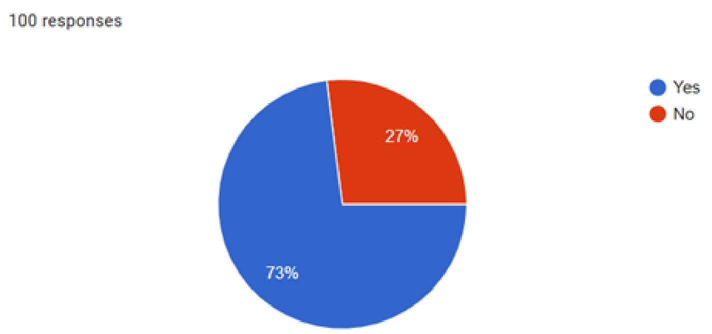

Figure 3-6: Willingness to buy electric scooters if the problems were solved.

The last question asked was for the people who answered the previous question with no and it was to state why they refused buying electric scooters even if solutions to the problems facing their spread was given. The answers were mostly that they don’t need it in their daily activities or because it isn’t safe and they will be scared to drive it.

In conclusion, the survey was instrumental in providing valuable insights into the needs and preferences of individuals regarding electric scooters. By analyzing the survey responses, we were able to gain a better understanding of the factors that can contribute to the wider adoption of electric scooters and the enhancement of user experiences. This information can be used to develop strategies to encourage the use of electric scooters and address the challenges that may hinder their widespread use. Ultimately, the findings of this survey can help to promote sustainable transportation options and contribute to a cleaner and more environmentally-friendly future.

Subsequently, an additional survey was conducted to obtain further insights into the usage of scooters in delivery services. The survey targeted delivery personnel from various chains and shops, who were asked a series of questions. The responses received were largely consistent across the board, providing a clear picture of the current trends and challenges in the delivery sector with regards to scooter use.

The primary objective of this survey was to ensure that the swappable battery station project would be beneficial to a wide range of individuals, particularly those who rely on gas-powered scooters for their daily work. By gathering information about the specific needs and concerns of delivery personnel, the project can be tailored to address their unique requirements and encourage the adoption of electric scooters. With the insights gained from this survey, we hope to promote sustainable and eco-friendly transportation options in the delivery industry, leading to a cleaner and healthier environment for all.

The first question asked was “what scooter do they usually use and what is the tank size?” Most of them replied with Dayun scooter which is popular brand of scooters that originated in China and its tank is 35 litres. It is known for producing affordable two-wheeled vehicles that are popular in many parts of the world.

The second question in the survey aimed to determine the average daily mileage of individuals who use scooters as their primary mode of transportation “how many kilos do you drive daily?” This information was sought to assess whether the cost of gasoline for these scooters was too high, and whether switching to electric scooters would be a more cost-effective option. Additionally, the survey aimed to determine whether the swappable battery station project would provide an incentive for individuals to switch to electric scooters, thereby contributing to the preservation of the environment. The responses to the question revealed that some individuals drive long distances due to their work requirements, while others drive shorter distances. The average daily mileage reported was approximately 150 km, indicating that individuals who rely on scooters for transportation cover a considerable distance on a daily basis. This information is useful in developing strategies to encourage the use of electric scooters, as it highlights the need for a reliable and efficient mode of transportation that can handle the daily mileage requirements of individuals. Overall, the survey's findings suggest that encouraging the adoption of electric scooters may provide a more cost-effective and environmentally-friendly solution for individuals who rely on scooters for transportation. The swappable battery station project can facilitate this transition by providing a convenient and accessible charging solution for electric scooter users, thereby promoting the wider adoption of sustainable transportation options.

The survey's third question inquired about the frequency of fueling the tank for individuals who use scooters as their primary mode of transportation “how many times do they fuel their tanks daily?” The purpose of this question was to gather information on the refueling habits of scooter users, which can be used to determine the potential cost savings associated with switching to electric scooters. The responses varied, with some individuals reporting that they refuel their tank once or twice daily, while others reported that they refuel depending on the distance they need to cover for deliveries. These findings suggest that the cost of gasoline can be a significant expense for individuals who rely on scooters for transportation. According to the delivery personnel who participated in the survey, they typically change their scooter's oil and filters twice a month. This routine maintenance adds an additional expense on top of the costs associated with fueling their scooters. Therefore, the adoption of electric scooters can provide a more cost-effective and sustainable alternative, as they require less frequent charging and can be refueled at a significantly lower cost compared to gasoline-powered scooters. Furthermore, the swappable battery station project can provide an additional incentive for individuals to switch to electric scooters, as it offers a convenient and accessible charging solution. This can further reduce the cost and inconvenience associated with refueling, leading to a more sustainable and efficient transportation option. In summary, the survey's findings on the frequency of refueling among scooter users highlight the potential cost savings and environmental benefits associated with the adoption of electric scooters. The swappable battery station project can provide a convenient and accessible charging solution, making it easier and more affordable for individuals to switch to electric scooters.

The final question of the survey aimed to gauge the willingness of individuals to purchase electric scooters if it meant spending less money than they currently do on gasoline-powered scooters. The responses to this question were overwhelmingly positive, with all participants expressing agreement. Additionally, respondents indicated that they would be more likely to purchase electric scooters if the issues of long charging times and slow speeds were addressed. The survey's findings suggest that the cost savings associated with electric scooters are a strong motivator for individuals to switch from gasoline-powered scooters. However, to encourage a wider adoption of electric scooters, it is crucial to address some of the challenges associated with their use, such as charging times and speed. This is where the swappable battery station project comes in, as it offers a solution to the charging problem by providing a convenient and accessible charging solution for electric scooter users.

Overall, the survey results indicate that the swappable battery station project has the potential to promote the use of electric scooters by addressing some of the challenges associated with their adoption. This, in turn, can lead to a more sustainable and affordable mode of transportation for individuals who rely on scooters for their daily transportation needs.

3.2 CASE STUDIES

Based on the conducted surveys and gathered data from scooter owners to estimate the average distance driven by scooter riders, which was approximately 100 km per day. Based on this information, we compared two scooter models, the Glide G1 and the SYM Fiddle 2. Both models had similar prices and shapes, with the Glide G1 priced at 60,000 EGP and the SYM Fiddle 2 priced at 52,500 EGP. In order to compare the cost of using these two models, we considered the price of the energy source for each. The price of 1 kWh of electricity was 1.45 EGP for the highest house electricity bundle, while 1 liter of fuel cost 10.25 EGP. We also factored in the additional expenses of oil changes every 1,000 km and filter changes every 2,000 km for the SYM Fiddle 2.

Based on these calculations, the estimated cost of using the SYM Fiddle 2 over a 10-year period for someone who drives 504 km per week would be 103,560 EGP. In comparison, the estimated cost of using the Glide G1 over the same period and distance would be 53,645 EGP. These findings highlight the potential cost savings of using an electric scooter like the Glide G1 over a gas-powered scooter like the SYM Fiddle 2. In addition to the financial benefits, electric scooters also have the advantage of being environmentally friendly, producing zero emissions and contributing to a cleaner, more sustainable future. Therefore, for individuals who frequently use their scooter for transportation, switching to an electric scooter like the Glide G1 can result in significant cost savings over time while also reducing their carbon footprint.

It is worth noting that the specific cost savings will depend on individual circumstances, such as the distance traveled, the frequency of usage, and the price of electricity and fuel in different locations. However, the comparison of the Glide G1 and the SYM Fiddle 2 provides a useful example of how electric scooters can be a cost-effective alternative to gas-powered scooters over the long term.

The table below compares the costs of using the Glide G1 electric scooter and the SYM Fiddle 2 gas-powered scooter over a period of 10 years, assuming a total distance of 259,200 km (504 km/week). The comparison is based on the energy source used by each scooter, the cost of that energy source, and the resulting cost of using each scooter for the given distance.

The first category is for someone who doesn't use their scooter much, riding only 1,050 km per month. By switching from the SYM Fiddle 2 to the Glide G1 electric scooter, they could save approximately 383.3 LE per month. This is because the cost of electricity to power the Glide G1 is significantly less than the cost of fuel to power the SYM Fiddle 2 over the same distance. Then, the second category is for someone who rides their scooter more frequently, covering a distance of 10,500 km per month. By switching to the Glide G1 electric scooter, they could save approximately 3,983 LE per month compared to using the SYM Fiddle 2. This is a significant cost saving and highlights the potential financial benefits of using an electric scooter for those who use their scooter frequently. Finally, the third category is for someone who uses their scooter extensively, riding 21,000 km per month. By switching to the Glide G1 electric scooter, they could save approximately 7,966 LE per month compared to using the SYM Fiddle 2. This is the highest potential cost savings of the three categories, as the cost of fuel for the SYM Fiddle 2 over such a long distance is much higher than the cost of electricity for the Glide G1.

Overall, the comparison shows that using an electric scooter like the Glide G1 can result in significant cost savings compared to using a gas-powered scooter like the SYM Fiddle 2. The cost savings increase with the frequency and distance of scooter use, making electric scooters a financially attractive option for those who rely on their scooter for daily transportation. Additionally, electric scooters have the added benefit of being environmentally friendly, producing zero emissions and contributing to a cleaner, more sustainable future.

Table 3-1: Comparison between G1 and SYM fiddle 2.

Point of comparison	*Glide G1*	*SYM Fiddle 2*
Price	60,000 EGP	52,500 EGP
Energy source	Electricity	Fuel
Energy source price	1 kWh = 1.45 EGP	1 liter fuel: 10.25 EGP
Cost for 210 Km	6.64 EGP	53.3 EGP
Cost per 1050 Km	33.2 EGP	266.5 EGP + 120 EGP + 30 EGP = 416.5 EGP
Cost per 10500 Km	332 EGP	2665 EGP + 1200 EGP + 300 EGP + 150 EGP = 4,315 EGP
Cost per 21000 Km	664 EGP	5330 EGP + 2400 EGP + 600 EGP + 300 EGP= 8,630 EGP
Cost for 10 years (259200 Km) (504 Km/week)	8195.66 EGP + 3*15150 EGP = 53,645.66 EGP	63960 EGP + 28800 EGP + 7200 EGP + 3600 EGP = 103,560 EGP

Another case study was conducted from a survey made for delivery guys to know which motorcycle is the most used for delivery applications which is the Dayun, the average distance travelled each day by those motorcycles and the different costs the delivery guys spend on their motorcycles. The comparison in this case study is between the Dayun motorcycle, which is commonly used for delivery purposes, and the Glide G1 electric scooter. The atudy aims to determine which vehicle is more cost-effective over a period of 10 years with 10 scooters per shop.

Based on the data from the survey, the average distance covered by delivery motorcycles was estimated and the cost of frequent oil and filter replacements were included in the calculations. Table 3-2 shows the results of the case study, where the price of 1 kWh of electricity used was 1.6 EGP for the highest electricity business bundle. The monthly cost for each vehicle model was calculated, with the Glide G1 costing 108 EGP per month and the Dayun costing 1,612 EGP per month when factoring in the cost of fuel, oil, and filter replacements. Then the cost for a shop having 10 scooters per month was estimated, with the Glide G1 costing 1,080 EGP for 10 scooters and the Dayun costing 16,120 EGP for the same number of scooters. Over a year, with the same 10 scooters covering 37,068 km, the cost of electricity for the Glide G1 was 12,960 EGP, while the cost of fuel, oil, and filter replacements for the Dayun was 193,440 EGP. Considering the lifespan of lithium-ion batteries to be approximately 10 years, the total cost of using 10 scooters over a period of 10 years was calculated. For the Glide G1, the total cost was 584,100 EGP, including the cost of replacing the battery three times over the 10-year period. In contrast, the total cost of using the Dayun motorcycle was 1,934,400 EGP, including the cost of fuel, oil, filter replacements, and engine overhauls.

Assuming a place with 40 branches, each having 10 scooters, the total cost of using the Glide G1 scooters for 10 years would be 47,364,000 EGP, while the total cost of using the Dayun motorcycles for the same period would be 91,776,000 EGP. This results in a cost savings of 44,412,000 EGP by using the Glide G1 scooters. The payback period for the investment in the Glide G1 scooters would be approximately 5 years, and the return on investment would be 200 percent. This shows that using electric scooters like the Glide G1 for delivery purposes can be a cost-effective and financially rewarding option compared to traditional gas-powered motorcycles like the Dayun.

Overall, the study highlights the potential cost savings of using electric scooters like the Glide G1 for delivery purposes, especially when considering the long-term costs of maintenance and fuel consumption. Furthermore, electric scooters have the added benefit of being environmentally friendly, producing zero emissions and contributing to a cleaner and more sustainable future.After conducting surveys and gathering data from delivery guys, we found that the Dayun motorcycle was the most commonly used motorcycle for delivery, despite the fact that it required frequent oil and filter replacements. We conducted a fleet management case study to compare the cost of using the Dayun motorcycle to the cost of using the Glide G1 electric scooter for delivery purposes.

Table 3-2: Fleet Management case study

Point of comparison	*Glide G1*	*Dayun*
Price	60,000 EGP	36,000 EGP
Energy storage	Li-ion Battery	Fuel tank 13 liter
Range price	1.2 kWh = 1.92 EGP	13-liter fuel = 133.25 EGP
Cost per month for 1 scooter	108 EGP	11320 EGP + 400 EGP + 80 EGP = 1612 EGP
Quantity	10	10
Cost per 1 month (3089 Km)	1080 EGP	113200 EGP + 4000 EGP + 800 EGP = 16,120 EGP
Cost per 1 year(37068 Km)	12,960 EGP	1358400 EGP + 480000 EGP + 96000 EGP = 193,140 EGP
Cost per 10 years	129600 EGP + (3*15150*10) EGP = 584,100 EGP	13584000 EGP + 480000 EGP + 960000 EGP = 1,934,400 EGP
For 40 branches	23,364,000 EGP	77,376,000 EGP
Cost of 400 Motorcycles	24,000,000 EGP	14,400,000 EGP
Total cost	47,364,000 EGP	91,776,000 EGP
Savings	+44,412,000 EGP	
Payback Period $=\frac{\text{Total cost}}{\text{Savings}}$	5 Years	
ROI $=\frac{\text{Savings}}{\text{Total cost}}$	200%	

C h a p t e r Four

4 BUSINESS PLAN

The A business plan is a comprehensive document that outlines the objectives, strategies, financial projections, and other essential details of a proposed business project. Its importance cannot be overstated, as it serves as a critical tool for entrepreneurs, investors, and stakeholders to evaluate the feasibility and potential success of a business venture.

A business plan is essential for several reasons. Firstly, it helps to clarify the business idea and develop a clear strategy for achieving the project's goals. The process of creating a business plan involves identifying the target market, analyzing competition, and determining the unique selling proposition of the product or service being offered. This process helps entrepreneurs to refine their business idea and develop a clear plan of action for achieving their objectives.

Secondly, a business plan provides a roadmap for the business's growth and development. It outlines the steps that need to be taken to achieve the project's goals, including the timeline and budget. A well-crafted business plan ensures that all stakeholders are aligned with the project's objectives and have a clear understanding of the steps that need to be taken to achieve success.

Thirdly, a business plan is crucial for attracting investors and securing funding. Investors want to see a well-crafted business plan that demonstrates the project's potential profitability and viability. A comprehensive business plan that includes detailed financial projections, market analysis, and a clear strategy for achieving the project's objectives is more likely to attract investors and secure funding.

Finally, a business plan serves as a benchmark for evaluating the business's performance and making adjustments as needed. By regularly reviewing and updating the business plan, entrepreneurs can ensure that the project is on track and make adjustments as needed to achieve success.

The swappable battery station project is an excellent example of the importance of having a business plan. The project targets places that have multiple branches and aims to provide a cost-effective and environmentally friendly solution for fleet management. The current unit contains four batteries, which cost a total of 85,000 LE. The project aims to replace gas-operated scooters with electric ones, which will significantly reduce the cost of maintenance and fuel.

Calculations show that a gas-operated scooter, such as the Dayun scooter, costs around 1,550 LE per month, including the cost of gas, oil, filters, and maintenance. However, if the scooter is replaced with an electric one, it will only cost 108 LE per month for electricity. This represents a significant cost saving for companies that rely heavily on their fleets for business operations.

To calculate the percentage of savings per month, the cost of electricity for the electric scooter was subtracted from the expenses of the gas-operated scooter. Therefore, 1,550 - 108 = 1,442 LE. This value was then divided by the expenses of gas scooters, resulting in a 93% savings from the cost of one gas-operated scooter. This demonstrates the potential cost savings that can be achieved by using electric scooters for fleet management.

The pricing of the project is based on the owners taking 33% from the savings percentage, which is equivalent to 551.5 LE per scooter per month. The place that will exchange all their gas-operated scooters with electric ones will have the other 60%, which is equal to 930 LE monthly. This pricing structure ensures that the project is financially sustainable and generates a return on investment for its owners.

Assuming a sale price of 511.5 LE per scooter per month and a total cost of the unit of 85,000 LE, the profit of the project, if the place that owns the unit has ten scooters, will be 5115 LE per month. This represents a significant return on investment and demonstrates the potential profitability of the swappable battery station project.

In conclusion, having a business plan is essential for any project or venture. It provides a clear roadmap for achieving the project's goals, helps to attract investors and secure funding, and serves as a benchmark for evaluating the business's performance. The swappable battery station project is an excellent example of the importance of having a business plan, as it demonstrates the potential cost savings and profitability that can be achieved by using electric scooters for fleet management.

4.1 SALES STRATEGY

It is important to note that calculating a client's savings can be a complex process that involves various factors such as operational costs, production efficiency, and market demand. Therefore, determining an appropriate pricing strategy that is both fair to the client and profitable for the owner requires a thorough understanding of the business landscape and the underlying economics. By conducting a detailed business plan, the client and owner can make informed decisions that maximize profitability while maintaining a competitive edge in the market.

Once the business plan was thoroughly analyzed and the projected profits and breakeven point were determined, the client's costing strategy was developed. It was previously stated that the complete unit, which includes four batteries, would be priced at 85000 LE. Based on the agreement between the owner and the client, the owner would receive 33% of the client's savings, resulting in a profit of 550 LE.

A subscription fee of 500 LE per scooter will be charged, in addition to a maintenance cost of 50 LE. This pricing structure results in a profit of 550 LE per scooter, equating to a total profit of 5500 LE for a location that owns ten delivery electric scooters. As part of the plan, the ten scooters will have their batteries exchanged at the unit. By agreeing to this plan, the client stands to save 10000 LE.

Another option for the clients in the project’s sales strategy is that they can order a special unit containing the number of battery chambers they need. But at this point the subscription and maintenance fee will be calculated per case as to ensure a balanced profiting for the owner and saving for the client.

It is worth noting that implementing a battery exchange program can result in significant cost savings for clients, as it eliminates the need to purchase and maintain expensive batteries. By striking the right balance, both parties can benefit from a mutually beneficial arrangement that promotes sustainability and efficiency.

C h a p t e r Five

5 SWAPPABLE BATTERY SYSTEM

The proposed solution chapter introduces the concept of swappable battery stations as a key solution to address the challenges of electric scooter usage. This chapter provides a comprehensive explanation of the system's operation, highlighting the iterative design process that led to its development. It presents a detailed schematic diagram showcasing the various components of the system and their interconnections, allowing for a clear understanding of the overall infrastructure. Additionally, a single line diagram is provided, outlining the electrical connections and pathways within the system. By presenting this proposed solution in a structured manner, this chapter aims to demonstrate the practicality and effectiveness of swappable battery stations in facilitating the seamless integration of electric scooters into urban environments.

5.1 OPERATION

The system consists of two simultaneous operations: one for the users and one for monitoring and protecting the unit. The user operation begins when a customer decides to swap their empty battery with a charged one. The operation requires various inputs, including a QR code scanner to open the designated chamber, a voltage sensor to check the battery's charge status, a limit switch to signal the locking mechanism when the door is closed, a timer to ensure the door is securely shut before locking, and a serial number verification to prevent incorrect battery placement.

When the user scans the QR code, the corresponding chamber door opens, allowing them to place the discharged battery inside. The system then performs three checks. First, it verifies the RFID serial number to ensure the correct battery is being replaced. If the serial number is valid, the system proceeds to check the voltage and the state of the limit switch to confirm that the battery is charging and the door is closed. If both conditions are met, the system waits for the timer to count down to 3 seconds before locking the door. Finally, another chamber door opens to allow the user to retrieve a fully charged battery. The user operation is illustrated in the provided flowchart.

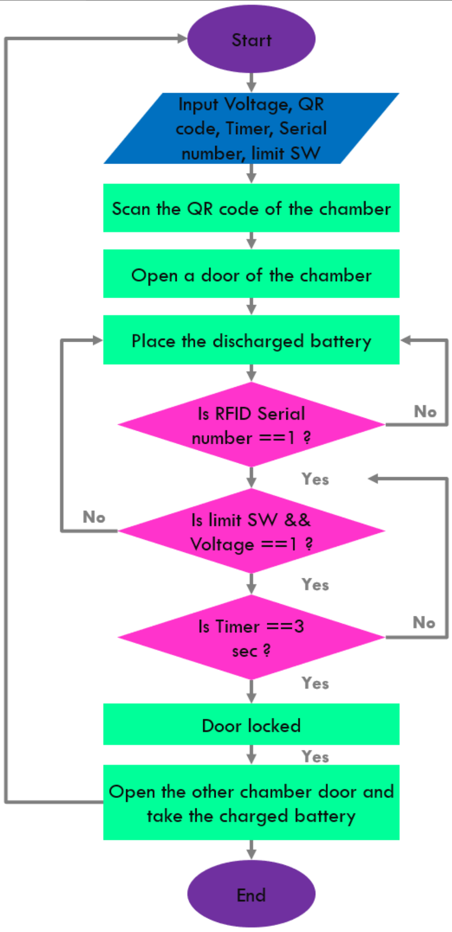

Figure 5-1: User operation.

The second operation of the system focuses on monitoring and protecting the unit. This operation runs continuously to ensure the safety of the batteries. The inputs for this operation include voltage and temperature sensors. While the batteries are being charged, the system constantly checks if the temperature exceeds or reaches 45°C. If the temperature threshold is reached, the system disconnects the batteries and identifies the underlying problem causing the temperature rise. This preventive measure aims to avoid overheating and potential damage to the batteries. The flowchart below illustrates the steps involved in this operation.

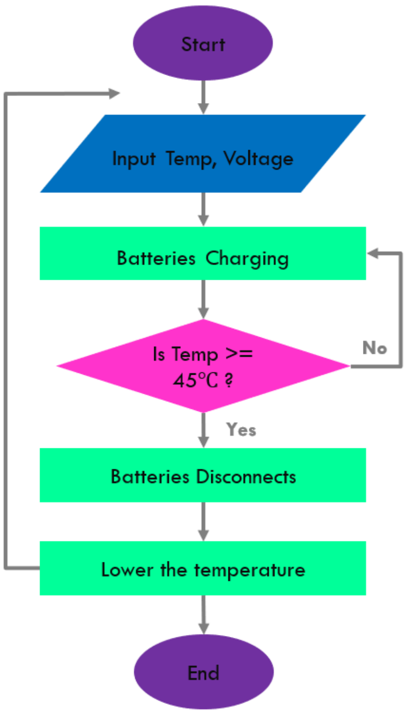

Figure 5-2: Monitoring operation.

5.2 ITERATIVE DESIGN

The iterative design process played a crucial role in the development of the battery swapping unit for this graduation project. The initial design featured a unit with four chambers, each containing two doors with two vertically positioned batteries. However, physical experimentation revealed that this design posed challenges in terms of handling the batteries due to their dimensions and weight. Additionally, the absence of a control unit to house the necessary controls, wires, and protection devices prompted the need for a second design iteration.

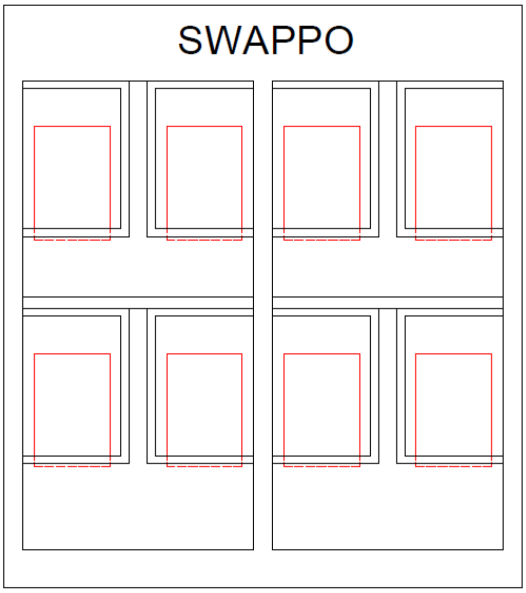

Figure 5-3: Initial Design.

In the second design iteration, the battery position was modified to a horizontal orientation, facilitating easier grabbing and placement of the batteries into the chambers. To reach this design with the accurate dimensions of the unit, all the dimensions of the battery were taken and spacings were made according to standards to ensure the batteries would fit and would have a proper ventilation. The figure below shows a schematic section and a schematic plan of the second design with the dimensions. The unit now comprised four vertically aligned chambers and each two chambers are connected together and interlocking such as A1 and A2. Moreover, a wire box was added and a control unit located at the back. While this design addressed the issues encountered in the previous iteration, it introduced a new concern of ensuring the battery's protection from scratches or impacts during handling, given the horizontal orientation with the battery handle on the side. A £D representation of this design is also viewed which was made using SketchUp Pro tool.

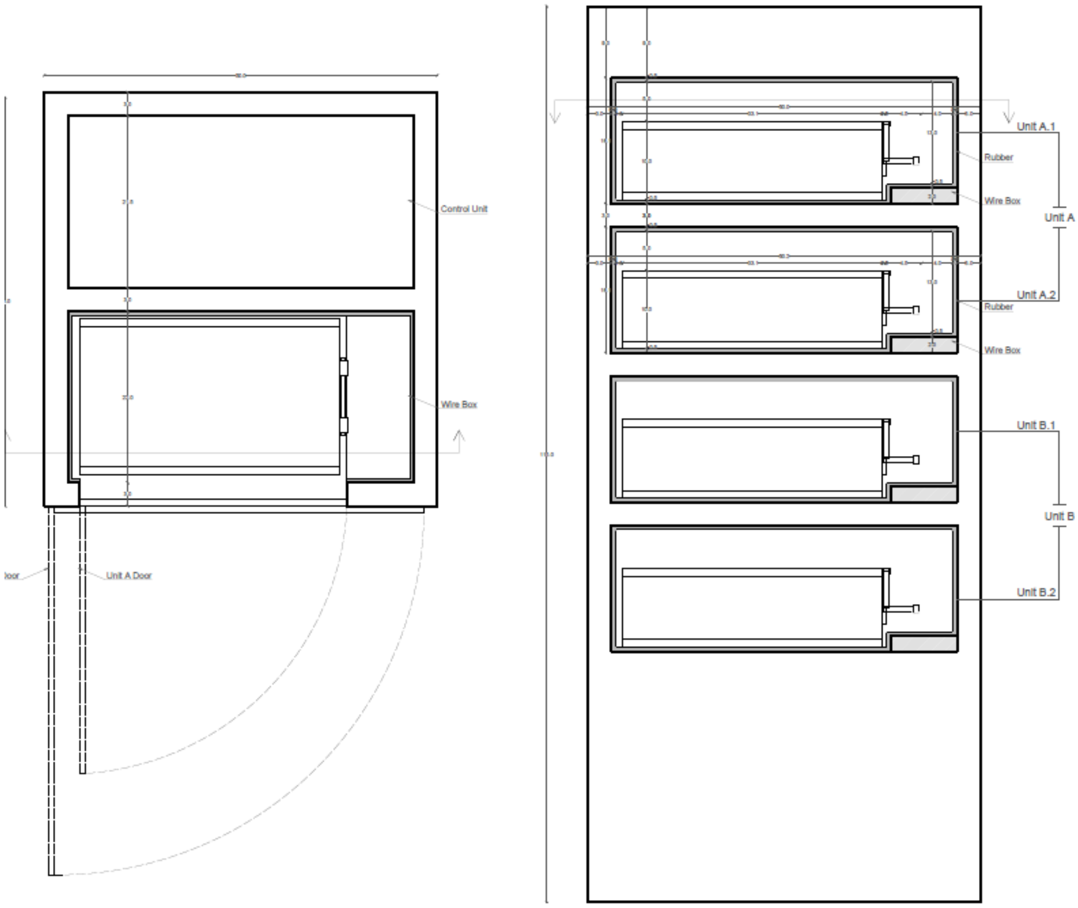

Figure 5-4: Second Design.

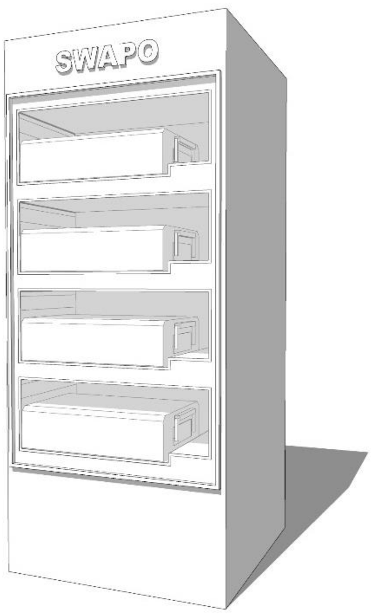

Figure 5-5: 3D Model of second Design.

To address this concern, a further modification was made in the subsequent design iteration. The batteries were still placed horizontally, but the handle was repositioned to face the user rather than being on the side. However, this adjustment resulted in an elongated unit, making it visually unappealing and consuming excessive space, thereby compromising ease of installation. This is shown in the following figure which illustrates a schematic section and plan of the third design.

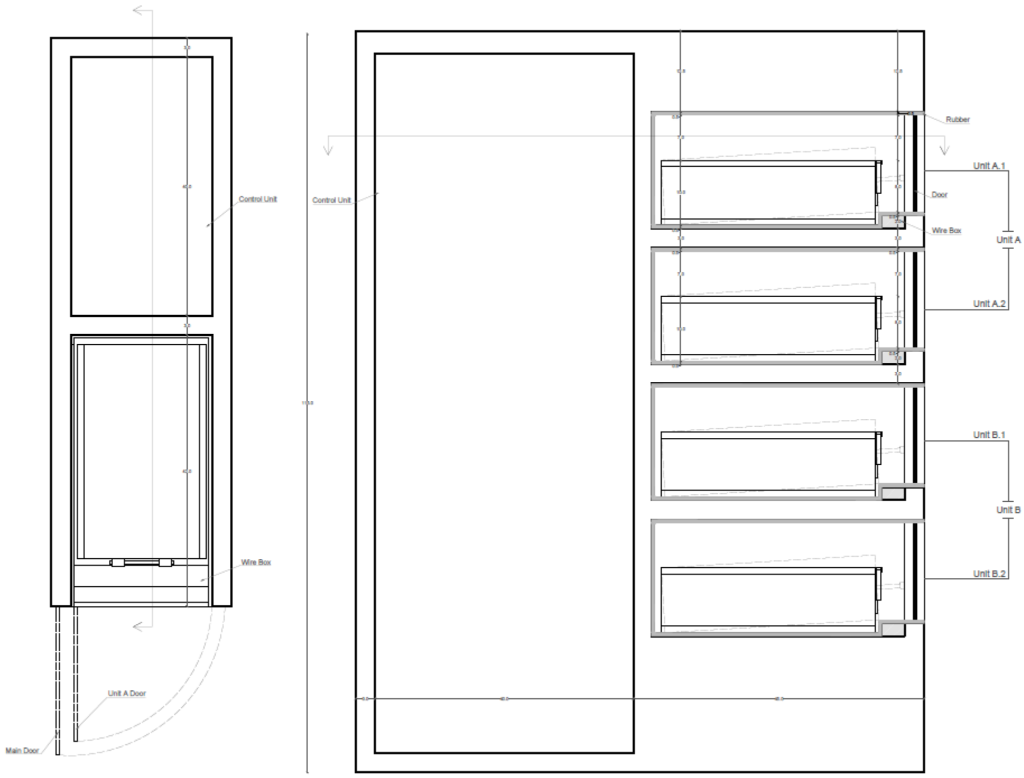

Figure 5-6: Third Design.

The final design emerged as a culmination of the iterative process, successfully resolving the challenges encountered in earlier iterations. As shown in the figures below, it features four vertically stacked chambers, with the batteries positioned horizontally and the handle facing the user. Additionally, the control unit is now situated alongside the chambers rather than behind them. This final design not only improves aesthetics but also enhances functionality, stability, ease of use, and accessibility.

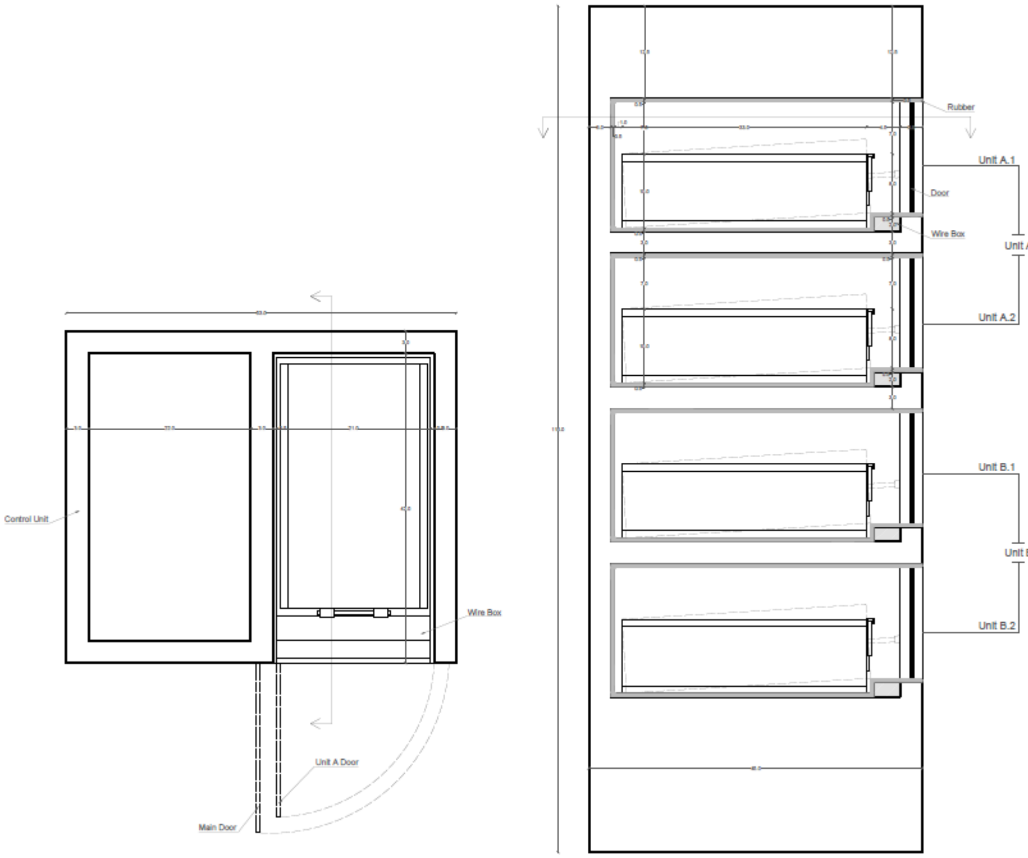

Figure 5-7: Final Design.

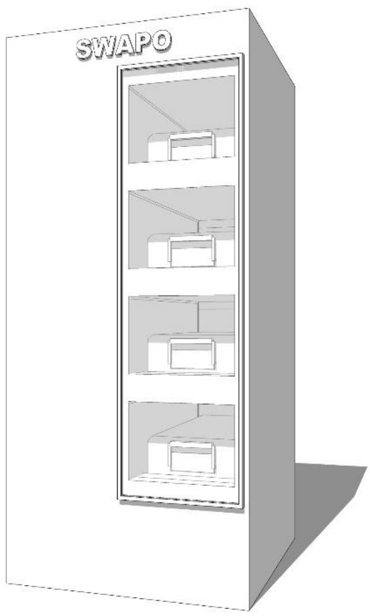

Figure 5-8: Final Design 3D.

Through the iterative design process, the battery swapping unit has undergone significant refinements, culminating in a final design that effectively addresses the project's objectives and requirements. The iterative approach allowed for continuous improvement and optimization, ensuring that the final design is efficient, user-friendly, visually appealing and suitable for the urban environment of Egypt.

5.3 SCHEMATIC DIGRAM

A schematic diagram is a visual representation of a system or process that uses symbols and lines to convey information. These diagrams can be found in various industries, from engineering and electronics to biology and chemistry. They are used to simplify complex systems and make them easier to understand. In order to effectively interpret a schematic diagram, it is important to understand the symbols and connections used in the diagram.

The schematic of the project was drawn on AutoCAD program to describe all the wiring of the components and their connection with each other. The arduino mega which is a microcontroller board based on the ATmega2560 chip is one of the largest and most powerful boards in the Arduino family, designed to provide more memory, input/output pins, and processing power than the standard Arduino Uno board. The Arduino Mega has 54 digital input/output pins, 16 analog inputs, 4 UARTs (hardware serial ports), a 16 MHz crystal oscillator, a USB connection, a power jack, an ICSP header, and a reset button. It can be programmed using the Arduino Integrated Development Environment (IDE) and is compatible with a wide range of sensors, actuators, and other electronic components is centered in the schematic diagram.

The arduino mega microcontroller is connected with the voltage sensors. Each voltage sensor has two battery inputs which are labelled I11 for the positive side and I12 for the negative side. The first digit after the letter I describes the number of the chamber, and the second digit if it is 1 then it shows the positive side and if it is 2 then it is the negative side. The voltage sensors also have three outputs which are the signal which is connected to the arduino, the VCC, and the GND. Their labels start with the letter V.

The limit switches have two outputs one of them is the NC and the other is GND. The limit switches are labelled with the letter L and also have two digits after the letter where the first digit stands for the chamber number and the second digit stands for the NC or the GND. Both NC and GND connect to the arduino microcontroller.

The DHT22 temperature sensors are connected to the arduino microcontroller as well and they are labelled with the letter D. the sensor has three outputs which are the signal, the VCC, and the GND.

The relay module is also connected to the arduino mega microcontroller. The relay module has eight relays. The first three relays are connected to the locks put on the doors of the chamber and they operate at 12v. The next three relays are connected to the AC side of the chargers of the batteries.

ETEK, EKM1-63S is the circuit breaker used to switch off the circuit if a short circuit ever occurs on the charger. Its specifications are 230/400 v, and holds up to 4500A, and 50/60 Hz. The circuit breaker is labelled U1 for the positive side and U2 for the negative side.

Lastly, there is an emergency button which opens all the doors of the chambers and it is labelled with the letter P.

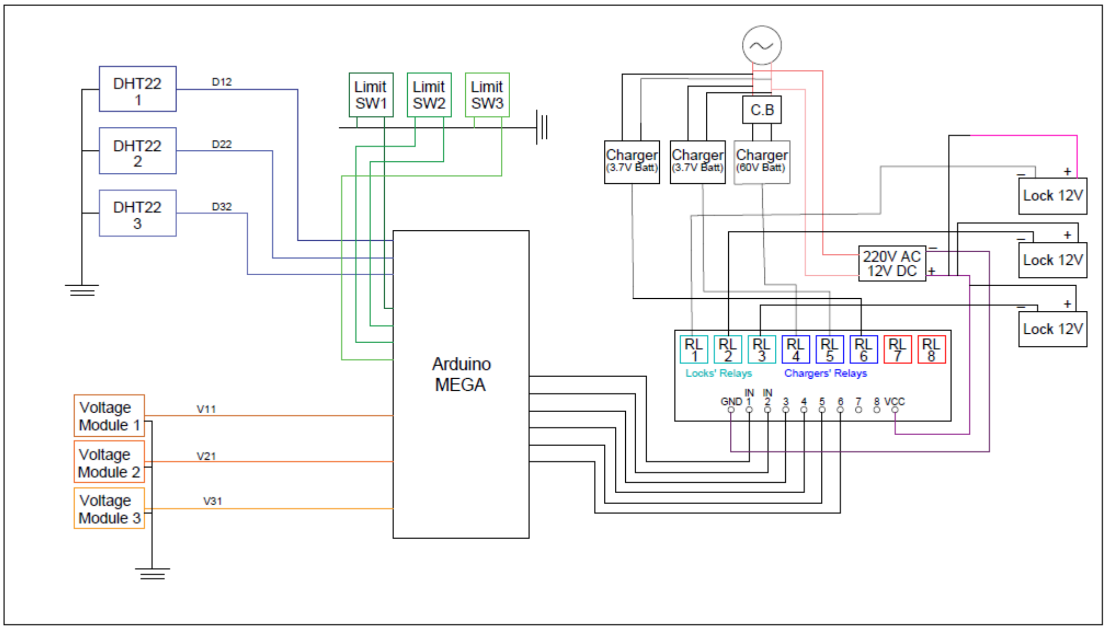

Figure 5-9: Schematic diagram.

Table 5-1: Wire labels

	Limit switch		Temperature sensor DHT22			Solenoid lock		Voltage sensor			Battery input	
	NC	GND	VCC	SIGNAL	GND	VCC	GND	SIGNAL	VCC	GND	VCC	GND
CHAMBER 1	L11	L12	D11	D12	D13	SL11	SL12	V11	V12	V13	I11	I12
CHAMBER 2	L21	L22	D21	D22	D23	SL21	SL22	V21	V22	V23	I21	I22
CHAMBER 3	L31	L32	D31	D32	D33	SL31	SL32	V31	V32	V33	I31	I32

Table 5-2: Circuit breaker and emergency button labels

Circuit breaker		Button	
VCC	GND	VCC	GND
U1	U2	P1	P2

Next, the team used the Proteus program to design the circuits for both the NodeMCU and the locks. It was necessary to ensure that the voltage is distributed on the locks equally because distributing the voltage evenly across the three locks helps ensure that each lock receives the same amount of power and operates correctly. If the voltage is not distributed evenly, one or more of the locks may not function properly, potentially compromising the security of the system. By designing the circuits to distribute the voltage evenly, the team was able to ensure that the locks functioned as intended and provided reliable security for the system.

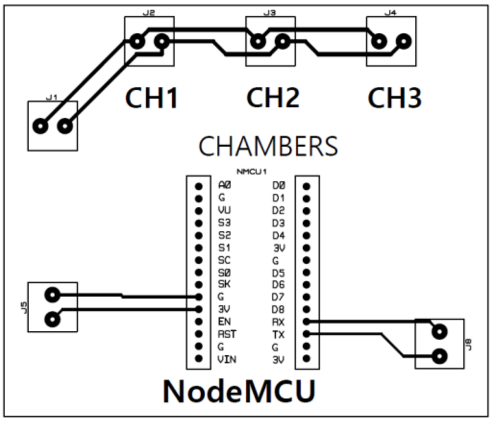

Figure 5-10: PCB layout.

5.4 SINGLE LINE DIAGRAM

SLD typically stands for "Single Line Diagram" in the context of electrical engineering. A Single Line Diagram is a simplified graphical representation of an electrical power system or a portion of it, where all the electrical components (such as generators, transformers, switchgear, breakers, and loads) are shown on a single line.

In an SLD, each component is represented by a standardized symbol, and the interconnections between them are shown as lines. The purpose of an SLD is to provide a clear and concise overview of the electrical system, showing the flow of power and the protection and control devices that are used to ensure safe and reliable operation.

An SLD can be used for a wide range of applications, such as in the design, construction, operation, and maintenance of electrical power systems. It is a valuable tool for engineers, technicians, and operators to understand the system and to communicate with each other effectively.

The single line diagram of the swappable batteries station project was drawn on AutoCAD program. The single line diagram represents the flow of electricity from the AC grid till it reaches the battery to charge it. Electricity is transmitted from the 220 volt and 50 Hz AC grid to a bus, which then distributes it to four miniature circuit breakers rated at 1P-32A. These circuit breakers are subsequently linked to the battery chargers, which require an AC input of 220 volts, 3 amperes, and 50 Hz, and provide a DC output of 71.4 volts and 4 amperes. The chargers are then linked to a rechargeable lithium-ion battery with a capacity of 60 volts and 24 ampere-hours. All of these components are connected using 1.5 mm flexible CU PVC cables from Elsewedy.

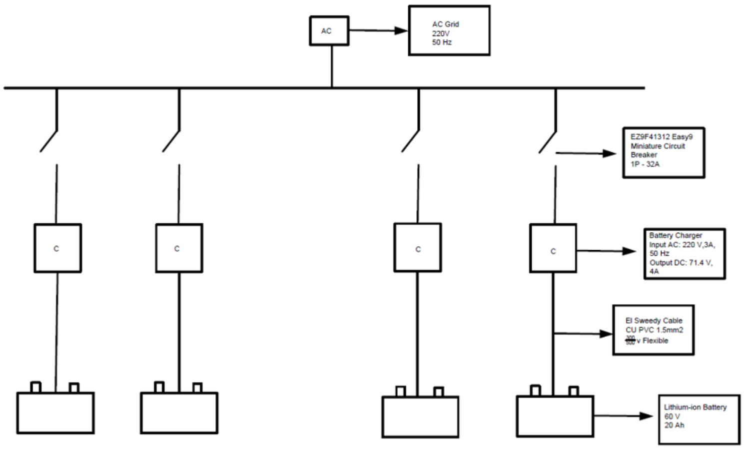

Figure 5-11: Single line diagram.

C h a p t e r Six

6 EXPERIMENTALAL IMPLEMENTATION

This chapter focuses on the practical application of the proposed solution through the development and testing of both a mini-model prototype and the main model. The chapter begins by describing the implementation of the mini-model, which serves as a preliminary prototype for the system. The mini-model is meticulously explained in detail, highlighting its design, functionality, and the specific components involved. The implementation process of the mini-model is thoroughly discussed, showcasing the steps taken to bring the prototype to life. Subsequently, the chapter delves into the main model, presenting its design and illustrating how it addresses the limitations and challenges identified in the mini-model. The implementation of the main model is elaborated upon, providing insights into the construction, integration, and configuration of its various components. Through this comprehensive examination of both the mini-model and the main model, this chapter demonstrates the progressive development and refinement of the proposed solution, laying the foundation for the subsequent evaluation and analysis of its performance.

6.1 MINI-MODEL

A mini model, also known as a small or compact model, can offer several benefits for a project, depending on the specific context and requirements. Some potential advantages of using a mini model are faster training and inference as mini models are typically smaller in size and have fewer parameters than larger models, which means they can be trained and run faster. Another advantage for the mini model is that it has lower computational and storage requirements. Mini models often require less computational power and storage space than larger models. Also, reduced complexity is a very important motive for building a mini model. As they can be simpler in design and easier to understand than larger models. This can make them more accessible to developers and researchers who may not have expertise in deep learning or related fields. As mini models have fewer parameters and a simpler architecture, mini models can be easier to interpret and analyze. This can be important for applications where explainability and transparency are crucial. Mini models also can be more easily deployed across different platforms and environments. It was easier to move it from a place to another to experiment it at different places.

In order to start building the mini model which showed the procedure of operation of the swappable battery station each component used in the mini model was tested alone in order to make sure that each and every component worked effectively. Firstly, the components tested were the power relay module. A relay power module is an electronic device that is used to control high-power electrical loads, such as motors, heaters, and lamps, using low-power control signals. It typically consists of a relay, which is an electrically operated switch that can be used to turn a high-power load on or off, and a control circuit that provides the low-power signals to operate the relay. Relay power modules can be designed to operate with different voltage and current levels, depending on the specific application requirements. They may also include additional features such as transient protection, overcurrent protection, and thermal protection to ensure safe and reliable operation. Overall, relay power modules provide a convenient and reliable way to control high-power loads using low-power signals.

Figure 6-1: Relay module.

Limit switches which are electromechanical devices that are used to detect the presence or absence of an object or to trigger an action when an object reaches a certain position or point were also tested. A limit switch typically consists of a movable actuator or lever that comes into contact with an object or surface, and a stationary housing that contains the switch mechanism. When the actuator is moved, it activates the switch, which can then send a signal to a controller or other device to initiate an action.

Figure 6-2: Limit switch.

Then LM35 temperature sensor was tested, although it worked efficiently it was changed by DHT temperature and humidity sensor. LM35 and DHT both are two types of sensors commonly used in electronics and IoT projects to measure temperature and humidity, respectively. The main differences between these two sensors are firstly, the type of measurement as LM35 is a temperature sensor only, while DHT is a combined temperature and humidity sensor. LM35 measures temperature in degrees Celsius (°C) with an accuracy of ±0.5°C, while DHT measures both temperature and relative humidity with an accuracy of ±2°C and ±5% RH, respectively. Another difference was the operating range where LM35 has an operating range of -55°C to 150°C, while DHT has an operating range of 0°C to 50°C for temperature and 20% to 90% for relative humidity. Power requirements of both sensors were different as well, as LM35 typically requires a power supply voltage of 4V to 30V and consumes very low power, while DHT requires a power supply voltage of 3V to 5V and consumes slightly more power than LM35. After deciding that DHT sensor was a better option for the mini model than the LM35 another comparison was done between DHT11 and DHT22. DHT11 and DHT22 are two types of temperature and humidity sensors so they both do the same functions. However, there were differences between the two such as the measurement range where DHT11 measures temperature between 0°C to 50°C with a ±2°C accuracy and relative humidity between 20% to 80% with a ±5% accuracy. DHT22 measures temperature between -40°C to 125°C with a ±0.5°C accuracy and relative humidity between 0% to 100% with a ±2% accuracy. Also, the response time as DHT11 has a slower response time of up to 2 seconds, while DHT22 has a faster response time of up to 0.5 seconds. The DHT11 has a lower sampling rate of once every 2 seconds, while DHT22 has a higher sampling rate of once every 0.5

seconds. Also, the power consumption was a key difference as DHT11 has a lower power consumption, requiring only 1mA during active measurement and 50uA during standby mode, while DHT22 requires 2.5mA during active measurement and 100uA during standby mode. Finally the price of DHT11 is more affordable than DHT22. But, the choice between DHT11 and DHT22 depends on the specific application requirements and the DHT22 fitted the requirements of the mini model better so it was chosen to be used.

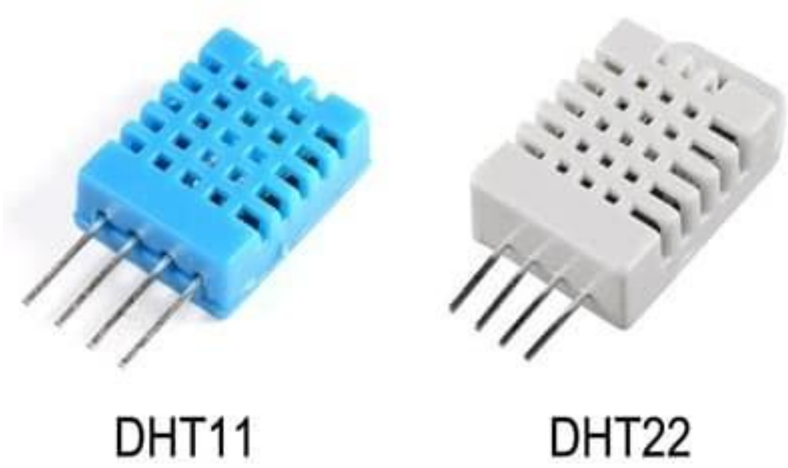

Figure 6-3: DHT sensors comparison.

Finally, the voltage sensor which is an electronic device that is used to measure the voltage level of an electrical circuit or system was tested. A voltage sensor converts the electrical voltage signal into a usable form that can be read by a microcontroller, analog-to-digital converter (ADC), or other digital device. Voltage sensors are commonly used in a variety of applications, including power management, battery monitoring, and renewable energy systems. They can be used to monitor the voltage of a battery to prevent overcharging or undercharging, or to monitor the output voltage of a solar panel or wind turbine to optimize power generation. Voltage sensors typically consist of a voltage divider circuit that reduces the input voltage to a safe level, and a conditioning circuit that amplifies and filters the signal to remove noise and interference. The output of the voltage sensor is typically a voltage level that is proportional to the input voltage, which can be read by a microcontroller or other digital device. Overall, a voltage sensor was a very important component in the mini model, as it provides a convenient and reliable way to measure the voltage level of an electrical circuit or system and can help prevent damage to equipment or improve system performance.

Figure 6-4: Voltage sensor.

After checking each component separately and making sure that it works effectively, small lithium ion batteries were used to imitate the large batteries of the scooters. The locks (relays) used in the mini model are normally open because in its normal condition there is no batteries in the chamber so there is no electricity running. Whenever a battery is connected and the limit switch is pushed for five seconds and the check of the battery's condition and temperature is done the relay gets a signal so the contact closes leading to electricity flowing to charge the battery that was placed in the chamber and the other lock opens (which means that the user is able to take the charged battery and put it in his scooter). During the implementation of the mini model the placement of the voltage sensor on the battery was a problem which was then solved by connecting the voltage sensor to the batteries by welding wires to the lithium ion batteries and connecting them to the sensor so it can read the voltages of the batteries. Another problem was that each relay had a separate gate drive circuit and many relays were used in the implementation of the mini model so a relay module which contains eight relays was used which eased the implementation of the mini model a lot.

6.2 UNIT MODEL

After implementing the mini model and making sure it worked correctly and satisfied all the requirements of the project the main unit was built. Many designs were drawn in order to pick the most suitable one and then the chosen design was sent to a factory to be done. The left side of the unit contains four chambers on top of each other. Each chamber is separated from the one above it with metal sheets containing holes to enable to movement of air inside the unit to prevent the overheating of the batteries inside the chambers. The chamber is covered with silicon sheets to prevent any damage that might occur to the batteries such as scratches that might lead to complete damage of batteries.

There is a lock on each chamber, and each lock is linked to a relay. Additionally, there is another lock on each chamber that is connected to the battery. This secondary lock serves to disconnect the battery from the electrical system if the temperature sensor detects a high reading. It also prevents the flow of electricity if the chamber is empty and does not contain a battery. To explain more this passage describes the system used for storing and charging the batteries. Where each chamber in the system contains a lock that is connected to a relay. This means that the system is designed to be controlled electronically, with the use of a microcontroller.

In addition to the lock connected to the relay, there is another lock on each chamber that serves a different purpose. This lock is connected to the battery and is designed to disconnect the battery from the electrical system if the temperature sensor reads a high value. This is likely a safety feature to prevent the battery from overheating or catching fire due to excessive heat. The secondary lock also prevents the flow of electricity if the chamber is empty and does not contain a battery. This is a safety feature to prevent damage to the system or to prevent a short circuit from occurring if the chamber is empty and the electrical contacts are exposed.

The voltage sensor which is supposed to read the voltage of the battery that will be put in the chamber is then connected to a voltage divider circuit. A voltage divider circuit is an electronic circuit that is used to divide a voltage into a lower voltage level that can be measured or used by a microcontroller, analog-to-digital converter (ADC), or other digital device. It consists of two resistors connected in series between the input voltage and ground, with the output voltage taken from the junction between the two resistors.

The voltage divider circuit works on the principle of Ohm's law, which states that the voltage across a resistor is proportional to the current flowing through it. In a voltage divider circuit, the resistors are chosen so that they have a specific ratio of resistance values. The output voltage is then determined by the ratio of the two resistors and the input voltage.

The formula for calculating the output voltage of a voltage divider circuit is Vout = Vin x (R2 / (R1 + R2)), where Vout is the output voltage, Vin is the input voltage, R1 is the resistance of the first resistor, and R2 is the resistance of the second resistor.

The values of the resistances used in the voltage divider were 10k ohms and 4k ohms as they were suitable and they give out the required value Vout which is the input to the the voltage sensor which is 25 volts.

Once the microcontroller in the unit receives a signal, the empty chamber without a battery is opened, and this synchronization is maintained with both the mobile application and the website. After a discharged battery is placed in the open chamber and the door is closed for five seconds, the limit switch is triggered and senses the battery's presence. The battery is then checked to ensure that it is 100% efficient, at the correct temperature, and is original. If the battery passes these checks, the chamber containing the more charged battery will open, allowing the user to remove it and install it in their scooter.

During the development of the mini model's code, it was originally designed to accommodate only two batteries. However, when the decision was made to update the code to support three batteries, additional work was required to ensure that the temperature sensor and relay, which function as the lock, were still operating efficiently and opening the empty chamber for the user to place their discharged battery.

Several problems arose during the code update process, including issues with the voltage sensor. The voltage sensor was reading the voltage of the charger, preventing the voltage of the batteries from being read. After analyzing the schematic of the charger and conducting further research, the code was modified to compare the voltages of two chambers instead of three, using Boolean and flag variables.

Another issue was encountered with the mechanical delay of the limit switches. However, this was resolved through coding adjustments. Overall, significant effort was required to address these issues and ensure that the updated code was functioning correctly and efficiently.

To complete the unit, each welded wire was covered with heat shrink tubing, and all connections made using rosettas were covered with terminals. The wires were then wrapped in spiral wrapping to protect users from electrical hazards, and secured to the walls of the unit to prevent access by the user. Omegas were positioned on the right side of each chamber, with chargers connected to 2 mm wires that were linked to relays and circuit breakers, and then to the power source.

An additional switch has been installed in the unit to grant access to all of the chambers in case of an emergency or for maintenance purposes, but only for authorized operators.

C h a p t e r Seven

7 CLOUD INTEGRATION

The cloud integration chapter explores the seamless integration of the proposed solution with cloud-based technologies, enabling users to effectively utilize the system. This chapter highlights the user experience by discussing the key components that facilitate interaction with the solution. Firstly, it examines the website, a central hub where users can access and control various functionalities of the system. The website serves as an intuitive interface, providing real-time data visualization, settings customization, and remote monitoring capabilities. Additionally, the chapter explores the mobile application, extending the accessibility of the solution to users on the go. The mobile application offers a user-friendly interface, empowering individuals to manage and monitor the system using their smartphones or tablets. Furthermore, the integration of Node-RED is showcased as a powerful tool that orchestrates the data flow between the system, website, and mobile application. It enables seamless communication and synchronization, ensuring that the user experiences a cohesive and integrated solution across all platforms. By examining the integration of these components, this chapter illuminates the user-centric design and functionality of the solution, ultimately enhancing the overall user experience and accessibility.

7.1 WEBSITE

The website developed is designed to help users locate the nearest battery station for their electric scooters. It uses geolocation technology to detect the user's current location and provides them with a list of nearby stations. The user can then select the station they prefer and view the available batteries. If they find a battery they want to reserve, they can proceed to the payment section, where they enter their name, mobile number, card number, expiry date, CVV, and the code of the desired battery. The website also offers directions to the selected station via Google Maps, making it easy for the user to find their way. Overall, the website provides a convenient and efficient way for electric scooter users to find and reserve batteries while on the go.

When the user opens the website, it provides him with some information including our vision which Is that we aim to revolutionize the utilization of vehicles and energy sources for more environmentally friendly urban riding. And we pledge to make urban riding safer, cleaner, and more affordable. The ideal option for anyone is swapo, which is our project's name as it allows you to save money and time. As well as explaining why do we use swappo, and highlighting the fact that the swappable batteries are a sustainable and cost-effective option for electric motorcycles. They reduce electronic waste, save money, and offer greater flexibility and mobility. With the growing availability of devices with swappable batteries, it is time to embrace this technology and make a positive impact on the environment. And then user can find their nearest swap which allows the website to detect the user's location and find the nearest station to them.

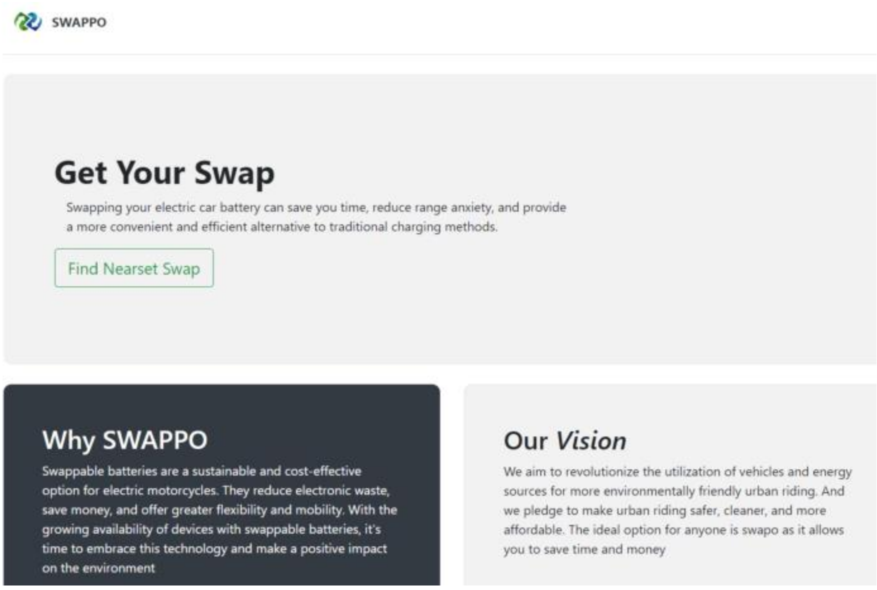

Figure 7-1: Website interface.

We experimented the website to make sure it is working properly by opening the website from sheikh Zayed and the two locations that had the stations were AAST which is located in smart village and is nearer to sheikh Zayed than the other station which was located in Doki. Instantly, the website chose the AAST station.

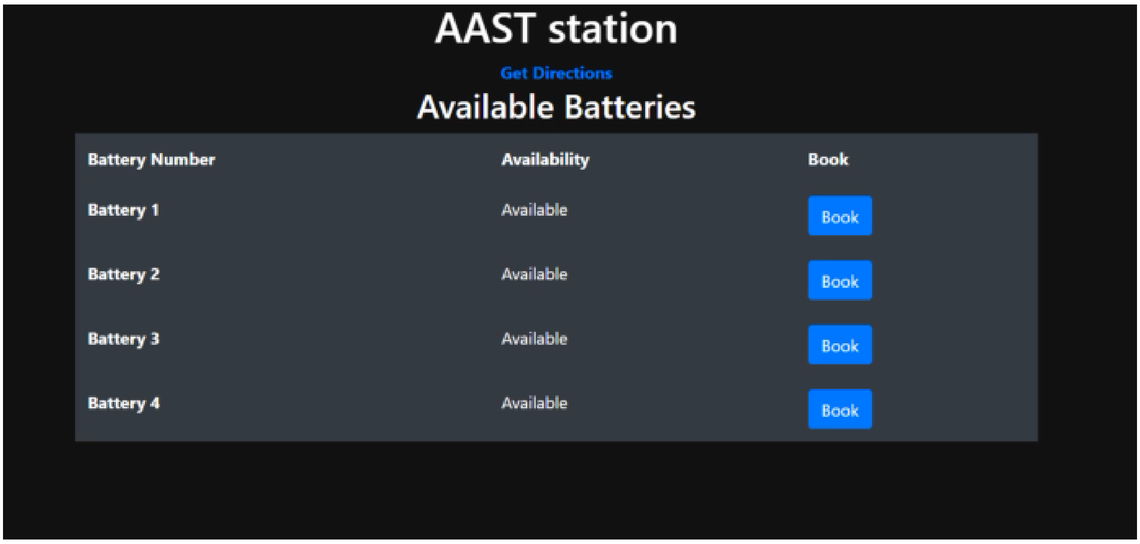

Figure 7-2: Booking page.

Then when the user chooses get directions it led him to the specific directions t the station with the help of google maps.

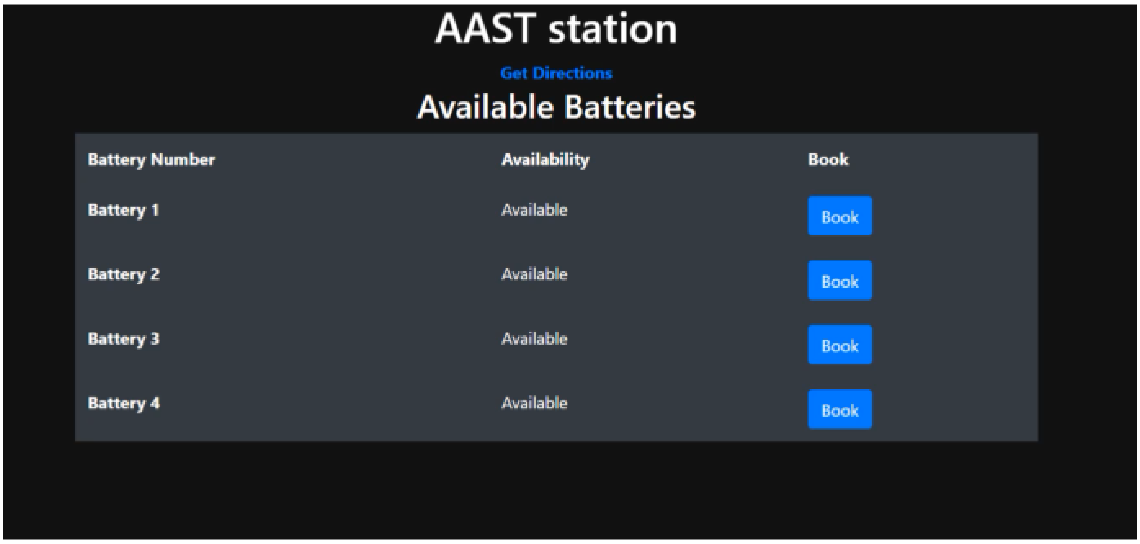

Figure 7-3: Booking process.

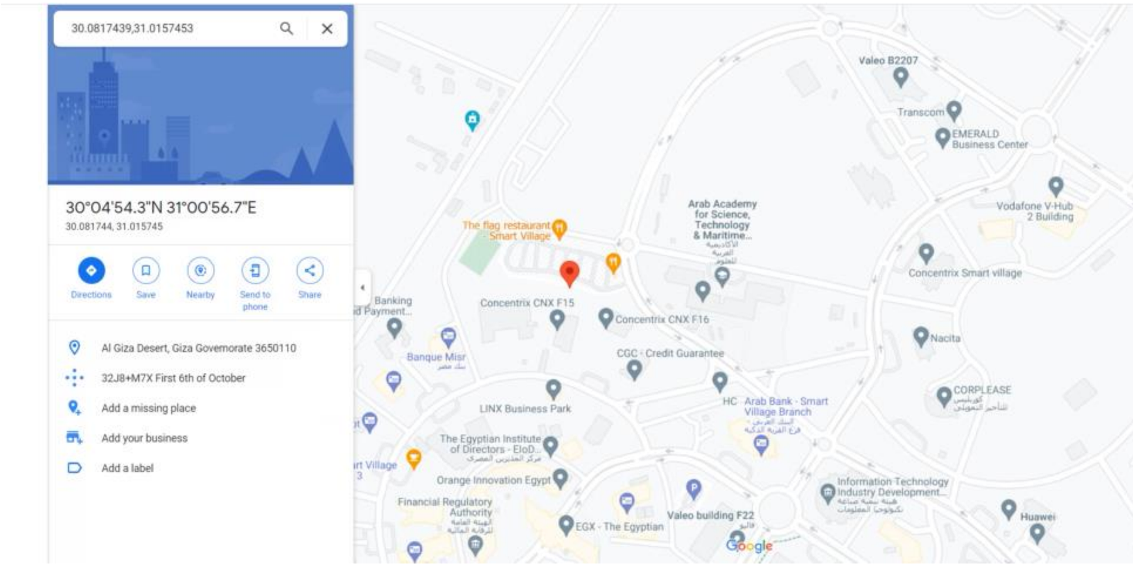

Figure 7-4: Station Location.

Then the user is provided with the number of available batteries and given the option to reserve one of them for a certain period of time until he reaches the station and is available to swap his discharged battery and place it in the station to be charged with an already charged one.

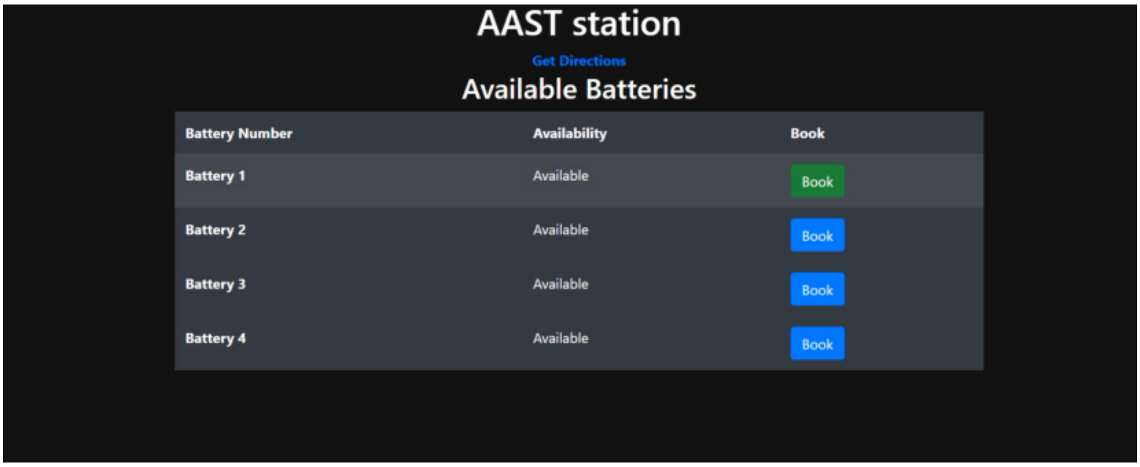

Figure 7-5: Choosing available battery.

Then the website directs the user to the final step which is the payment step where the user enters their name, mobile number, card number, expiry date, and cvv. Then by submitting payment the user will be given a code which allows the access to the chamber that contains the charged battery.

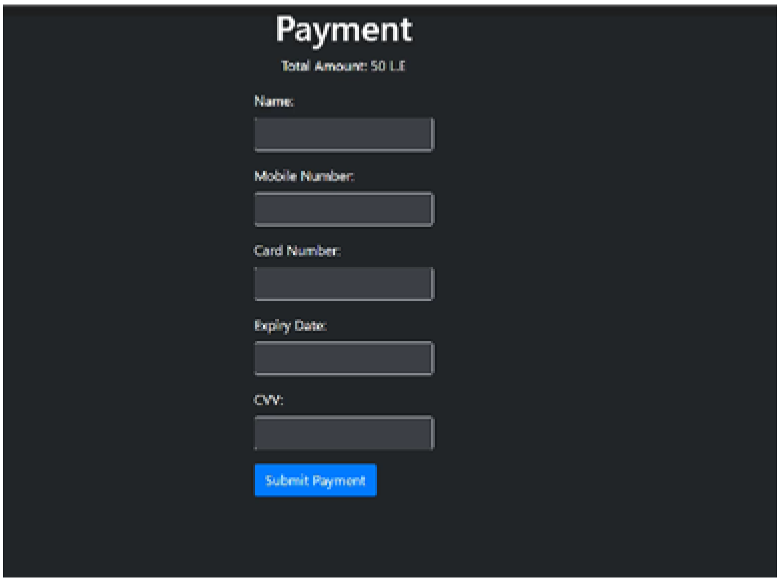

Figure 7-6: Payment page.

7.2 MOBILE APPLICATION

Mobile applications have revolutionized the way we interact with technology and have made our lives simpler and more convenient than ever before. With just a few taps on a screen, we can access a wealth of information and services. Mobile apps have made it possible for us to complete tasks quickly and efficiently, without the need for physical interaction or lengthy processes. Mobile apps are constantly evolving, with new features and functionalities being added all the time, making our lives even more convenient. Overall, mobile apps have become an integral part of our daily lives, enabling us to accomplish tasks with ease and efficiency, and enhancing our overall productivity and quality of life.

The mobile application for this project is designed to provide a convenient and user-friendly way for users to access the project's features and functionalities on their smartphones or tablets. The app's interface is intuitive and visually appealing, with easy-to-navigate menus and buttons that allow users to perform tasks quickly and efficiently. The app is designed to be responsive and compatible with a wide range of mobile devices, ensuring that users can access the project's resources from anywhere, at any time. The app includes features such as real-time notifications, personalized user profiles, and a comprehensive search function to help users find the information they need.

The primary objective of our app, SWAPO, is to alleviate concerns about running out of battery power while on route to a destination, while also promoting the use of eco-friendly energy sources and reducing traffic congestion by encouraging the use of E-scooters. The name SWAPO is derived from the concept of swapping out depleted batteries for fully charged ones to ensure that drivers have enough energy to reach their destinations.

Once the application is opened the user finds a map that contains all the different locations of the swappable batteries stations.

Figure 7-7: Application welcome page

The user then can choose to reserve a battery by clicking the button that says reserve a battery in the upper right part of the page.

Figure 7-8: Reserving a battery.

Once the user clicks on reserve a battery a page is opened which requires his first name, last name, email, and password.

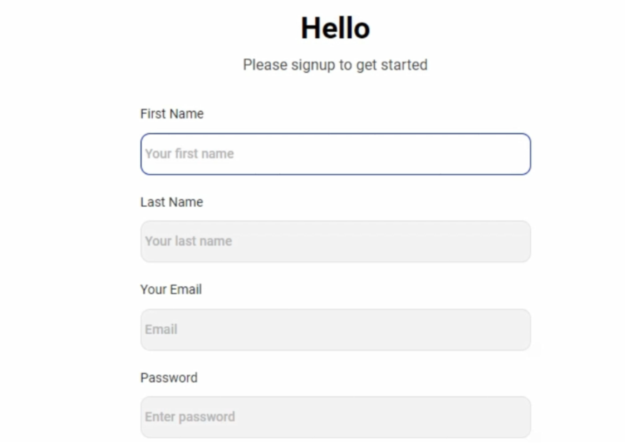

Figure 7-9: Info page.

When the user enters the required information the mobile application will ask for the ID number, this feature is made to prevent stealing, and by implementing this feature we ensure preventing any suspicious activity as well as ensuring integrity of the mobile application. The mobile application also requires a profile picture for the user in order to achieve maximum security.

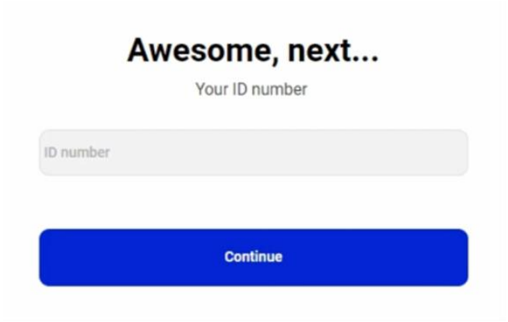

Figure 7-10: ID number.

Figure 7-11: Face ID.

Then the stations near the user's location appear and he is able to choose the nearest to him as well as reserving a battery in it for a certain period till he reaches it.

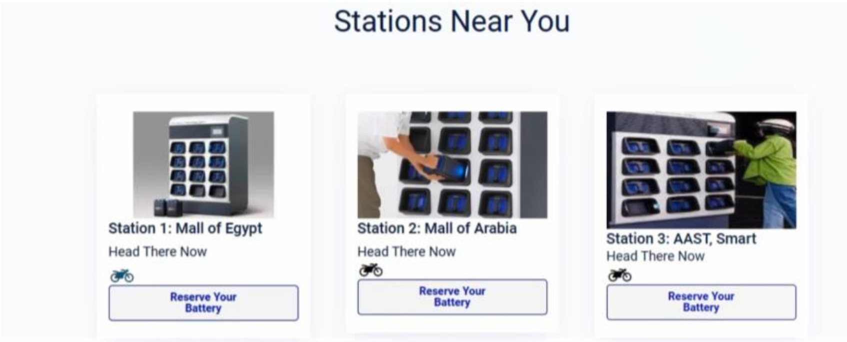

Figure 7-12: Locating the nearest station.

Lastly the user will be navigated to the payment page where there are several option for payment. A QR code will appear for the user after he is done with the payment part that he will later on use in opening one of the chambers of our station in order to take the charged battery and replace it with the empty one.

7.3 NODE-RED

Node-RED is an open-source visual programming tool that enables users to easily connect and automate devices, systems, and data flows. It provides a browser-based interface with a drag-and-drop approach, allowing users to create and deploy applications by connecting pre-built nodes that represent various functions and devices.

One of the key advantages of Node-RED is its simplicity and ease of use. It provides a visual interface that eliminates the need for traditional coding, making it accessible to users with little or no programming experience. The visual nature of Node-RED enables quick prototyping and rapid development of applications. Node-RED's versatility and extensibility are additional advantages. It offers a wide range of nodes that can interact with different devices, protocols, and APIs, making it compatible with various systems and technologies. Moreover, users can create their own custom nodes to extend the functionality of Node-RED to meet specific requirements.

Moreover, Node-RED finds applications in diverse domains. It is commonly used for IoT (Internet of Things) applications, where it facilitates the integration and management of sensors, actuators, and data streams. Node-RED can also be employed for workflow automation, data processing, and real-time analytics. Its visual interface and extensive library of nodes make it suitable for building home automation systems, smart cities applications, industrial automation, and more.

Overall, Node-RED's user-friendly interface, extensibility, and compatibility with various systems make it a powerful tool for connecting and automating devices and data flows in a wide range of applications. Its simplicity and versatility have contributed to its popularity among developers and IoT enthusiasts. All the above mentioned reasons are why Node-RED was chosen as the programming tool for IoT application in this project.

6.3.1 Getting Started with Node-RED

To open Node-RED and access its interface, these steps should be followed:

1. Install Node-RED: Node-RED can be installed on your local machine, a server, or a cloud-based platform. You can check the official Node-RED website for detailed installation instructions specific to your operating system or deployment environment.
2. Launch Node-RED: Once Node-RED is installed, you can launch it by running the appropriate command or clicking on the Node-RED application icon.
3. Access the Node-RED Editor: Node-RED provides a browser-based editor interface, which you can access by opening a web browser and entering the URL provided during the installation process. Typically, the URL is http://localhost:1880/ or it could be accessed by opening the Node-RED command prompt and initializing Node-RED to launch it and generate the IP address for the browser-based editor interface.

The Node-RED interface is shown in figure 7-13 and discussed below:

- Flow Editor: The main part of the Node-RED interface is the Flow Editor. It consists of a workspace where you build your flows by connecting nodes together. Nodes represent different functions, devices, or data sources. You can drag and drop nodes from the palette onto the workspace and connect them to define the flow of data and actions.
- Palette: The Palette is located on the left side of the interface. It contains a wide range of pre-built nodes organized into categories. You can search for specific

nodes or browse through the categories to find the desired functionality. Nodes can be expanded to reveal their configuration options.

- Flow Workspace: The Flow Workspace is the central area where you create and edit your flows. It provides a canvas where you can arrange nodes, draw connections between them, and define the flow logic. You can zoom in/out and navigate the workspace to manage complex flows.
- Node Configuration: When you select a node on the workspace, its configuration panel appears on the right side of the interface. This panel allows you to set the properties and parameters of the selected node, such as input/output settings, connection details, and other specific options.
- Deploy and Debug: At the top-right corner of the interface, you'll find the deploy button, which is used to deploy your flows and make them active. During deployment, Node-RED checks the flow for errors and starts executing the defined actions. The debug sidebar displays messages and information generated during the execution, helping you monitor and troubleshoot the flow.
- Additional Functionality: Node-RED offers various additional features and tools. For example, you can import/export flows, manage nodes and their configurations, install new nodes from the Node-RED library, and use the built-in text editor for more advanced programming.

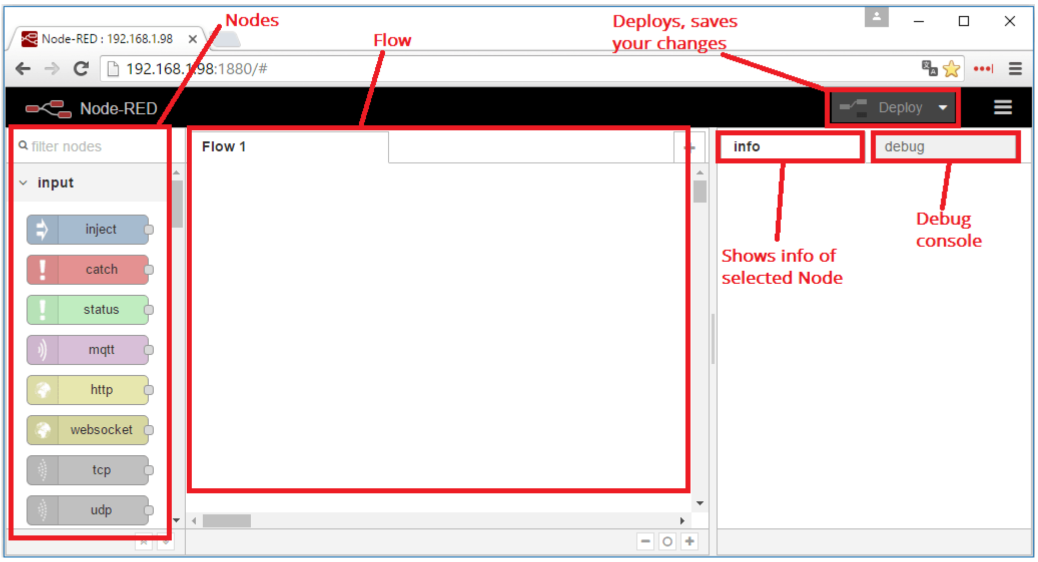

Figure 7-13: Node-RED interface.

6.3.2 IoT in Swappable Batteries Station using Node-RED

This section will discuss the integration of data gathered from the unit with the website and mobile application: how the data entered by the user through the website or the mobile application will be received by the unit to take the required action, and how the unit processes and sends data from sensors and users back to the app or website to be updated frequently and for monitoring.

In order for this integration to happen, the data must be saved on and sent through a server. The server used in this project is the Mosquitto MQTT Broker which is an MQTT broker that facilitates communication between devices and applications using the MQTT protocol. Mosquitto implements the MQTT (Message Queuing Telemetry Transport) protocol, which is a lightweight publish-subscribe messaging protocol designed for efficient communication in constrained environments, such as IoT devices. MQTT follows a publish-subscribe model, where devices can publish messages to specific topics, and other devices can subscribe to those topics to receive the published messages. Where the topics are the voltage, temperature and humidity. These are topics which will be subscribed to. While the code entered by the user to open the unit chamber door to get the battery is a message published toanother topic.

To keep it simple, in this project, the Arduino Mega which runs the whole system and all the sensors, relays and switches are connected to, reads the incoming data from the sensors. Then sends this data to a NodeMCU which in this project is used to provide Wi-Fi to enable access to the server and the Node-RED. The NodeMCU receives the data from the Arduino through serial communication. When provided power, the NodeMCU attempts to connect to the Wi-Fi and to the broker, if the attempt was successful and a serial communication was available to allow flowing of data from Arduino uno if the broker was subscribed to the topic this data is being published to, then data will be published to the broker and anyone could see it through MQTTLens or an application called MyMQTT. In the Node-RED, once data is published to the broker on a certain topic, then using the mqtt out node in Node-RED and subscribing to the same topic where data is being published to, then the Node-RED is now receiving the data from the sensors connected to the Arduino Mega. Then some nodes such as gauges or charts were used to display the output data. So, when the topic that holds the data from the DHT22 sensor is subscribed to and the data is displayed on a gauge in the Node-RED dashboard, the flow and the output becomes as shown in the figures below.

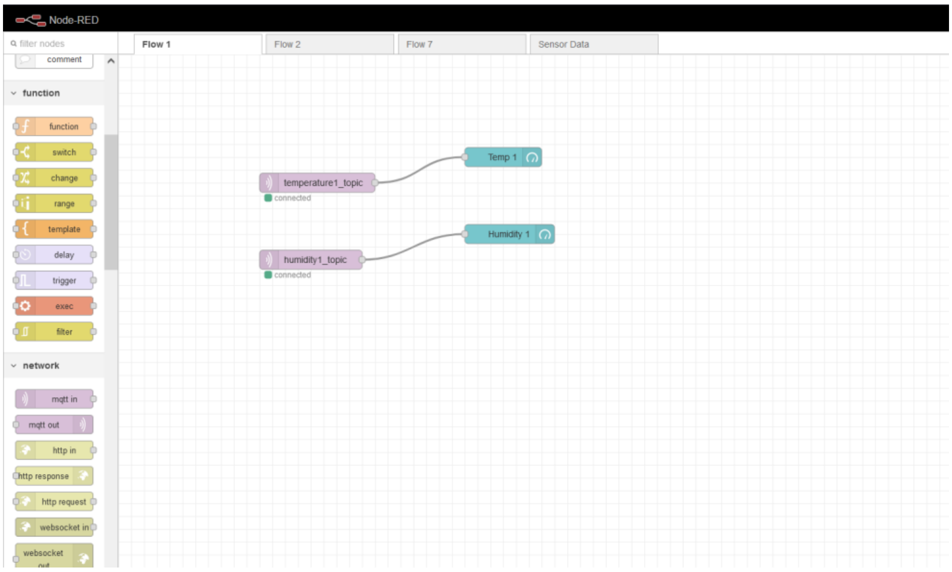

Figure 7-14: Node-RED flow.

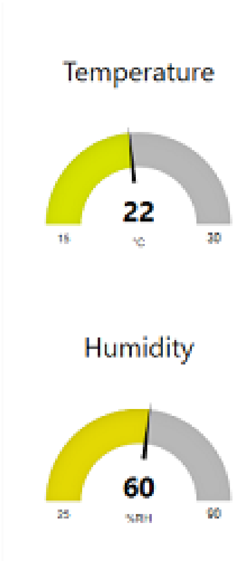

Figure 7-15: Output on Node-RED dashboard.

The NodeMCU code is the code responsible for this whole publishing or subscribing to data and this is how it operates in this project. The program commences by establishing a wireless network connection to a specified network, employing the ESP8266 module integrated within the NodeMCU. This connection is achieved by supplying the network name (SSID) and password as credentials. Upon successful connection, the program displays the assigned IP address of the NodeMCU, signifying a successful network connection. To facilitate communication with the MQTT broker, the MQTT protocol is utilized. The program connects to the MQTT broker, which is identified by an IP address or domain name, employing the PubSubClient library. It configures the MQTT client, defines a callback function to handle incoming messages, and establishes a connection to the MQTT broker.

Within the main program loop, the code continually checks for incoming data from the Arduino Mega. If data is available, it is read until a newline character is encountered. The received data adheres to a specific format, exemplified as "1:25.5,50.0,3.3,1;2:26.0,55.0,3.4,0;3:26.5,60.0,3.5,1;". This data encompasses temperature, humidity, voltage, and relay state values for each chamber. The program processes the received data by extracting the relevant values associated with each chamber. This is accomplished by utilizing auxiliary functions tailored to retrieve values based on the sensor ID. For instance, the function getValue() extracts temperature and humidity values, while getVoltageValue() retrieves the voltage value, and getRelayStateValue() retrieves the relay state. Based on the received command, the program publishes the extracted data pertaining to each chamber to the MQTT broker. A string is constructed, containing the sensor data in a predefined format, such as "T1:25.5 H1:50.0 V1:3.3 Charger1:1". This constructed string is then published to the corresponding MQTT topic for each chamber, such as "chamber/1", "chamber/2", and "chamber/3". In the event of a disconnection from the MQTT broker, the program endeavors to reconnect at regular intervals until a successful reconnection is established. Upon successful reconnection, the program publishes an announcement to the MQTT broker, indicating a successful connection, and resubscribes to the "chamber/command" topic to receive commands from the broker.

In summary, the code exemplifies a comprehensive system that enables the transfer of data from an Arduino Mega to an MQTT broker via a NodeMCU. This configuration permits remote monitoring and control of chambers equipped with relays, DHT sensors, and voltage sensors. By utilizing the MQTT protocol, a lightweight and efficient communication mechanism for IoT applications, seamless integration and data exchange are facilitated.

C h a p t e r Eight

8 ENVIRONMENTAL AND ECONOMIC IMPACT

The environment is of paramount importance to all life on earth. It provides the essential resources that sustain us, including clean air, water, and food, as well as the raw materials we use to create the goods and services we rely on. The environment is also home to a diverse array of plants and animals, many of which are essential to the ecosystem's balance and function. Moreover, a healthy environment is crucial for our physical and mental well-being, as exposure to pollution and other environmental hazards can have detrimental effects on our health. The environment is a finite resource, and we must protect it to ensure that future generations can also enjoy its benefits. Therefore, it is critically important to prioritize the protection and preservation of the environment in all decision-making processes.

The environment plays a crucial role in sustaining life on earth in a variety of ways. For example, the air we breathe is essential for our survival, and the earth's atmosphere acts as a natural filter, regulating the amount of harmful pollutants that enter it. Moreover, the environment plays a crucial role in shaping our physical and mental health. Exposure to air and water pollution, as well as other environmental hazards, has been linked to a variety of health problems, including respiratory disease, cancer, and neurological disorders. On the other hand, spending time in nature has been shown to have a range of positive effects on our mental health, reducing stress and anxiety and improving mood. Finally, it is essential to recognize that the environment is a finite resource. The earth's natural resources are not infinite, and we must use them responsibly to ensure that they are available for future generations. Protecting the environment is an investment in our collective future, and it is essential to prioritize sustainability in all decision-making processes to ensure that we leave a healthy planet for our children and grandchildren.

Our project has a very good environmental impact. It aims for a healthy, not polluted environment. As a start it works on reducing carbon dioxide emissions. Using electric scooters can significantly reduce CO_2 emissions and help to mitigate the impacts of climate change. Unlike traditional gas-powered scooters, electric scooters produce zero emissions during operation. This means that they do not contribute to local air pollution, which is a significant problem in many urban areas. By using electric scooters instead of gas-powered vehicles, we can reduce the amount of harmful pollutants released into the atmosphere, including carbon monoxide, nitrogen oxides, and particulate matter.

Electric scooters also offer a more energy-efficient mode of transportation. Compared to gas-powered vehicles, electric scooters require significantly less energy to operate, which means that they have a lower carbon footprint. Additionally, electric scooters can be charged using renewable energy sources like solar or wind power, further reducing their environmental impact.

Another benefit of using electric scooters is that they can help to reduce traffic congestion. In urban areas, traffic congestion is a significant contributor to air pollution and greenhouse gas emissions. By using electric scooters, we can reduce the number of cars on the road, which can help to alleviate traffic congestion and reduce emissions.

Electric scooters have the potential to reduce traffic, fuel consumption, and noise, making them an attractive option for urban transportation. Firstly, electric scooters are small and nimble, which means they can navigate through traffic more easily than larger vehicles. This ability to maneuver through traffic can help to reduce congestion on busy roads and improve overall traffic flow.

Secondly, electric scooters have low energy consumption compared to traditional gas-powered vehicles. In fact, electric scooters require only a fraction of the energy needed to operate a gas-powered vehicle, making them much more efficient. This means that we can reduce our reliance on fossil fuels and lower our carbon footprint by using electric scooters instead of gas-powered vehicles.

Finally, electric scooters are significantly quieter than gas-powered vehicles. This is because they use electric motors instead of internal combustion engines, which produce noise and vibration. By reducing noise pollution, electric scooters offer a more enjoyable and peaceful mode of transportation for both riders and pedestrians.

In conclusion, electric scooters have the potential to reduce traffic congestion, fuel consumption, and noise pollution in urban areas. By providing a more efficient and sustainable mode of transportation, electric scooters can help to create a cleaner, quieter, and more livable city environment. As such, electric scooters offer a promising solution to the challenges of urban transportation, and their use is likely to become increasingly popular in the coming years.

Furthermore, lining with government directions is one of the goals that swappable battery station project aims for as it was said that during the upcoming years all the scooters used in delivery will be electrical in order to reduce carbon dioxide emissions to help decrease the percentage of pollution and global warming.

Swappable battery stations can promote the usage of electric scooters in several ways. Firstly, they can help to address the issue of range anxiety, which is a common concern among electric scooter users. Range anxiety refers to the fear of running out of battery power and being stranded in an inconvenient location. By providing swappable battery stations in strategic locations, electric scooter users can easily swap out their depleted battery for a fully charged one, thereby extending the range of their electric scooter and reducing the fear of running out of power.

Secondly, swappable battery stations can help to reduce the cost of electric scooters. One of the main barriers to the adoption of electric scooters is the high cost of the battery. By providing swappable battery stations, electric scooter manufacturers can offer a more affordable option for consumers. Instead of purchasing an expensive battery, users can simply pay for the use of a fully charged battery at a swappable battery station. This can help to make electric scooters more accessible to a wider range of consumers.

Thirdly, swappable battery stations can help to reduce the environmental impact of electric scooters. By providing a system for recycling and reusing batteries, swappable battery stations can help to reduce the amount of waste generated by electric scooters. This can help to reduce the overall carbon footprint of electric scooters and make them a more sustainable mode of transportation.

In summary, swappable battery stations can promote the usage of electric scooters by addressing the issue of range anxiety, reducing the cost of electric scooters, and reducing the environmental impact of electric scooters. As such, swappable battery stations offer a promising solution to the challenges of electric scooter adoption and are likely to become increasingly popular in the coming years.

C h a p t e r Nine

9 CONCLUSION AND FUTURE WORK

9.1 CONCLUSION

The Swappable Battery Station project has the potential to revolutionize the transportation industry. By providing a convenient and accessible charging solution for electric scooter users, the project can promote the adoption of electric scooters as a sustainable and cost-effective mode of transportation.

The project has several important benefits. It can significantly reduce the operating costs of electric scooters. Currently, one of the main barriers to the adoption of electric scooters is their high upfront cost. However, the SBS project can reduce the cost of ownership by providing a convenient and affordable charging solution for electric scooter users. This can make electric scooters a more attractive option for individuals who rely on scooters for their daily commute or deliveries.

Secondly, the project can promote the wider adoption of sustainable transportation options. Electric scooters have significantly lower emissions than gasoline-powered scooters, making them a more environmentally-friendly transportation option. By promoting the adoption of electric scooters, the SBS project can contribute to a cleaner and more sustainable environment.

Thirdly, the project can improve the convenience and accessibility of electric scooters. One of the main challenges associated with electric scooters is the need for frequent charging. However, the SBS project can eliminate this challenge by providing a convenient and accessible charging solution for electric scooter users. This can improve the overall convenience and accessibility of electric scooters, making them a more attractive transportation option for individuals.

Finally, the project can promote the development of new technologies and innovations in the transportation industry. As the adoption of electric scooters increases, there will be a greater demand for new technologies and innovations that improve the performance, efficiency, and convenience of electric scooters. The SBS project can serve as a catalyst for the development of new technologies and innovations in the transportation industry, leading to a more sustainable and efficient transportation system.

In conclusion, the Swappable Battery Station project is an important initiative that has the potential to revolutionize the transportation industry. By providing a convenient and accessible charging solution for electric scooter users, the project can promote the adoption of electric scooters as a sustainable and cost-effective mode of transportation. The project can significantly reduce the operating costs of electric scooters, promote sustainable transportation options, improve the convenience and accessibility of electric scooters, and promote the development of new technologies and innovations in the transportation industry. Overall, the Swappable Battery Station project is an important step towards a more sustainable and efficient transportation system.

9.2 FUTURE WORK

Looking ahead, the Swappable Battery Station project has several plans for future developments. These include updates to the mobile application and website to enhance user-friendliness and make it easier for users to access the charging service. Additionally, the charging unit will be expanded to accommodate a larger number of battery chambers, allowing for simultaneous charging of multiple batteries.

As the project gains momentum and popularity, it is expected that the idea of swappable battery stations will spread in many more locations. This increased visibility and accessibility will further promote the adoption of electric scooters as a sustainable transportation option.

The updates to the mobile application and website will focus on improving the user experience and making the process of locating and accessing the charging service more seamless. This is crucial to ensuring that the project is accessible and convenient for all users, regardless of their level of technological proficiency. By enhancing the user-friendliness of the platform, the project team hopes to encourage more individuals to adopt electric scooters and take advantage of the cost savings and environmental benefits they offer.

Expanding the charging unit's capacity is also an important step towards promoting the adoption of electric scooters on a larger scale. By accommodating a larger number of battery chambers, the charging unit can serve more users simultaneously, thereby reducing wait times and increasing accessibility. This is particularly important in densely populated areas where there is a high demand for transportation options.

Finally, as the idea of swappable battery stations gains more visibility and popularity, it is expected that the project will spread in many more locations. This will make it easier for individuals to access the charging service and promote the adoption of electric scooters as a sustainable and cost-effective mode of transportation. The project team is committed to expanding the reach of the project and ensuring that swappable battery stations are available in as many locations as possible.

In summary, the Swappable Battery Station project has several exciting plans for future developments. These include updates to the mobile application and website, expansion of the charging unit's capacity, and spreading the idea of swappable battery stations to more locations. By enhancing the accessibility and convenience of the charging service, the project team hopes to promote the adoption of electric scooters as a sustainable and cost-effective mode of transportation.

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APPENDIX A – CODES

APPENDIX A.1 – ARDUINO CODE

```
#include <DHT.h>

#include <SoftwareSerial.h>

#define DHTPIN1 28    // DHT temperature sensor 1

#define DHTPIN2 29    // DHT temperature sensor 2

#define DHTPIN3 30    // DHT temperature sensor 3

#define DHTTYPE DHT22   // DHT22 temperature sensor is used

#define VOLTAGE_SENSOR_PIN1 A0   // Voltage sensor 1 is connected to analog pin A0

#define VOLTAGE_SENSOR_PIN2 A1   // Voltage sensor 2 is connected to analog pin A1

#define VOLTAGE_SENSOR_PIN2 A2   // Voltage sensor 3 is connected to analog pin A2

const int relay1Pin = 32;        // Pin number for relay 1

const int relay2Pin = 33;        // Pin number for relay 2

const int relay3Pin = 34;        // Pin number for relay 3

#define RELAY_charging1 35   // Relay module 4 is connected to digital pin 37

#define RELAY_charging2 36   // Relay module 5 is connected to digital pin 38

#define RELAY_charging3 37   // Relay module 6 is connected to digital pin 39

int unitflag1 = 0 ;

int unitflag2 = 0 ;

int unitflag3 = 0 ;

int unit1 = 0 ;

int unit2 = 0;

int unit3 = 0 ;

int timer1 = 0;
```

```
int timer2 = 0;

int timer3 = 0;

int limitSwitch1State = 0;     // Variable to store the state of limit switch 1

int limitSwitch2State = 0;     // Variable to store the state of limit switch 2

int limitSwitch3State = 0;     // Variable to store the state of limit switch 3

const int chargedbattery ;

const int buttonPin = 24; //the pin number of the push button10

int buttonState = 0; //variable for reading the button status

// Define the pins for the limit switches

const int limitSwitch1Pin = 45;

const int limitSwitch2Pin = 46;

const int limitSwitch3Pin = 47;

DHT dht1(DHTPIN1, DHTTYPE);

DHT dht2(DHTPIN2, DHTTYPE);

DHT dht3(DHTPIN3, DHTTYPE);

int offset = 20; // set the correction offset value


unsigned long startTime1 = 0;    // Variable to store the starting time of limit switch 1 press

unsigned long startTime2 = 0;    // Variable to store the starting time of limit switch 2 press

unsigned long startTime3 = 0;    // Variable to store the starting time of limit switch 3 press


unsigned long relay1StartTime = 0; // Variable to store the starting time of relay 1 activation

unsigned long relay2StartTime = 0; // Variable to store the starting time of relay 2 activation

unsigned long relay3StartTime = 0; // Variable to store the starting time of relay 3 activation
```

}

```
const unsigned long relayActiveDuration = 3000; // Duration for which relays will be active (in milliseconds)


SoftwareSerial mySerial(5, 6); // RX, TX pins for SoftwareSerial

int control;


void setup() {

  pinMode(relay1Pin, OUTPUT); // Set relay 1 pin as output

  pinMode(relay2Pin, OUTPUT); // Set relay 2 pin as output

  pinMode(relay3Pin, OUTPUT); // Set relay 3 pin as output

  pinMode(RELAY_charging1, OUTPUT);

  pinMode(RELAY_charging2, OUTPUT);

  pinMode(RELAY_charging3, OUTPUT);

  pinMode(buttonPin, INPUT); //initialize the push button pin as an input

  // Set the limit switch pins as inputs with internal pull-up resistors

  pinMode(limitSwitch1Pin, INPUT);

  pinMode(limitSwitch2Pin, INPUT);

  pinMode(limitSwitch3Pin, INPUT);


  // Start the serial communication

  Serial.begin(9600);  // Serial Monitor for debugging

  mySerial.begin(9600); // SoftwareSerial for NodeMCU

  dht1.begin();

  dht2.begin();

  dht3.begin();

}
```

```
void loop() {
  float h1 = dht1.readHumidity();
  float t1 = dht1.readTemperature();
  float h2 = dht2.readHumidity();
  float t2 = dht2.readTemperature();
  float h3 = dht3.readHumidity();
  float t3 = dht3.readTemperature();

  //float v1 = analogRead(VOLTAGE_SENSOR_PIN1) * (5.0 / 1023.0);   // Convert analog value to voltage
  //float v2 = analogRead(VOLTAGE_SENSOR_PIN2) * (5.0 / 1023.0);   // Convert analog value to voltage
  int volt1 = analogRead(A0);// read the input
  double v1 = map(volt1, 0, 1023, 0, 2500) + offset; // map 0-1023 to 0-2500 and add correction offset
  v1 /= 100; // divide by 100 to get the decimal values

  int volt2 = analogRead(A1);// read the input
  double v2 = map(volt2, 0, 1023, 0, 2500) + offset; // map 0-1023 to 0-2500 and add correction offset
  v2 /= 100; // divide by 100 to get the decimal values

  int volt3 = analogRead(A2);// read the input
  double v3 = map(volt3, 0, 1023, 0, 2500) + offset; // map 0-1023 to 0-2500 and add correction offset
  v3 /= 100; // divide by 100 to get the decimal values
  // Read the state of limit switch 1
  /*int limitSwitch1State = digitalRead(limitSwitch1Pin);
```

```
  // Invert the state of limit switch 1 since internal pull-up resistors are used
  // limitSwitch1State = !limitSwitch1State;
  // Read the state of limit switch 2
  int limitSwitch2State = digitalRead(limitSwitch2Pin);
  // Invert the state of limit switch 2 since internal pull-up resistors are used
  //limitSwitch2State = !limitSwitch2State;
  // Read the state of limit switch 3
  int limitSwitch3State = digitalRead(limitSwitch3Pin);
  // Invert the state of limit switch 3 since internal pull-up resistors are used
  //  limitSwitch3State = !limitSwitch3State;*/

if (isnan(h1) || isnan(t1) || isnan(h2) || isnan(t2)) {
  Serial.println("Failed to read from DHT sensor!");
  return;
}
buttonState = digitalRead(buttonPin); //read the state of the push button
 if (buttonState == HIGH) { //if the button is pressed open all doors
   digitalWrite(relay1Pin, HIGH);
   digitalWrite(relay2Pin, HIGH);
   digitalWrite(relay3Pin, HIGH);
   Serial.println("open unit all");
   delay(15000);
   digitalWrite(relay1Pin, LOW);
   digitalWrite(relay2Pin, LOW);
   digitalWrite(relay3Pin, LOW);
  }
```

```
if (mySerial.available()) {
    control = mySerial.read();  // Read the incoming data from NodeMCU


delay(50);
}


 if ( control == 1)    //Connection to from MQTT broker to start the system
 {
   if (v1 < 1) { // OPen Chamber 1
     digitalWrite(relay1Pin, HIGH);
     digitalWrite(RELAY_charging1, LOW); // normaly closed relay on charger is now closed and current flowing to charger
     Serial.println("unit1 opened");
     unit1 = 1;
     // Check if limit switch 1 is pressed for 5 seconds
     /*if (limitSwitch1State == HIGH) {
        timer1 = 0;
        Serial.println("LIMIT 1 PRESSED");
        while (timer1 < 2000 && limitSwitch1State == LOW) {
          limitSwitch1State = digitalRead(limitSwitch1Pin);
          //limitSwitch1State = !limitSwitch1State;
          delay(10);
          timer1 += 10;
       }}*/
     while (unit1 == 1) {
       limitSwitch1State = digitalRead(limitSwitch1Pin);
       delay(100);
```

```
if (limitSwitch1State == HIGH) { // Limit switch 1 is pressed
  Serial.println("Limit switch 1 is pressed");
  if (startTime1 == 0) {
    startTime1 = millis(); // Record the starting time of the limit switch press
  }

  if (millis() - startTime1 >= 500) { // Check if 2 seconds have elapsed
    if (relay1StartTime == 0) {
      relay1StartTime = millis();
      volt1 = analogRead(A0);// read the input
      v1 = map(volt1, 0, 1023, 0, 2500) + offset; // map 0-1023 to 0-2500 and add correction offset
      v1 /= 100; // divide by 100 to get the decimal values
      if (v1 > 1) {
        Serial.println("Chamber 1 Closed");
        digitalWrite(relay1Pin, LOW);// Trip relay 3 if voltage is more than 2 volts for voltage sensor 1
        if (v2 >= v3) {

          Serial.println("Chamber 2 open");
          digitalWrite(RELAY_charging2, HIGH); // normaly closed relay on charger is now open and no current flowing to empty chamber
          digitalWrite(relay2Pin, HIGH);
          delay(3000);
          digitalWrite(relay2Pin, LOW);
          Serial.println("Chamber 2 closed");
          relay1StartTime = 0; // Reset the relay 1 starting time
          unitflag2 = 0;
```

```
          unitflag1 = 1;

          unit1 = 0;

         }

         else if (v3 > v2) {


          Serial.println("Chamber 3 open");

          digitalWrite(RELAY_charging3, HIGH); // normaly closed relay on charger is now open and no current flowing to empty chamber

          digitalWrite(relay3Pin, HIGH);

          delay(3000);

          digitalWrite(relay3Pin, LOW);

          Serial.println("Chamber 3 closed");

          relay1StartTime = 0; // Reset the relay 1 starting time

          unitflag3 = 0;

          unitflag1 = 1;

          unit1 = 0;


         }

        }

       }

      }

     } else {

      startTime1 = 0; // Reset the starting time if the limit switch is released

     }

    }

   }
```

```
  else if (v2 < 1 ) { // OPen Chamber 2
    digitalWrite(relay2Pin, HIGH);
    digitalWrite(RELAY_charging2, LOW); // normaly closed relay on charger is now closed and current flowing to charger
    Serial.println("unit opened 2");
    unit2 = 1;
    /*if (limitSwitch2State == HIGH) {
      timer2 = 0;
      while (timer2 < 2000 && limitSwitch2State == LOW) {
        limitSwitch2State = digitalRead(limitSwitch2Pin);
        //  limitSwitch2State = !limitSwitch2State;
        delay(10);
        timer2 += 10;
      }

      }*/
    while (unit2 == 1) {
      limitSwitch2State = digitalRead(limitSwitch2Pin);
      delay(100);
      if (limitSwitch2State == HIGH) { // Limit switch 2 is pressed
        Serial.println("Limit switch 2 is pressed");
        if (startTime2 == 0) {
          startTime2 = millis(); // Record the starting time of the limit switch press
        }

        if (millis() - startTime2 >= 500) { // Check if 2 seconds have elapsed
          if (relay2StartTime == 0) {
```

```
relay2StartTime = millis(); // Record the starting time of relay 2 activation

volt2 = analogRead(A1);// read the input

v2 = map(volt2, 0, 1023, 0, 2500) + offset; // map 0-1023 to 0-2500 and add correction offset

v2 /= 100; // divide by 100 to get the decimal values



if ( v2 > 1) {

  Serial.println("Chamber 2 Closed");

  digitalWrite(relay2Pin, LOW);// Trip relay 3 if voltage is more than 2 volts for voltage sensor 1

  if (v1 >= v3) {



    digitalWrite(RELAY_charging1, HIGH); // normaly closed relay on charger is now open and no current flowing to empty chamber

    digitalWrite(relay1Pin, HIGH);

    Serial.println("Chamber 1 open");

    delay(3000);

    digitalWrite(relay1Pin, LOW);

    Serial.println("Chamber 1 closed");

    relay1StartTime = 0; // Reset the relay 1 starting time

    unitflag1 = 0;

    unitflag2 = 1;

    unit2 = 0;

  }

  else if (v3 > v1) {



    digitalWrite(RELAY_charging3, HIGH); // normaly closed relay on charger is now open and no current flowing to empty chamber

    digitalWrite(relay3Pin, HIGH);
```

```
          Serial.println("Chamber 3 open");
          delay(3000);
          digitalWrite(relay3Pin, LOW);
          Serial.println("Chamber 3 closed");
          relay1StartTime = 0; // Reset the relay 1 starting time
          unitflag3 = 0;
          unitflag2 = 1;
          unit2 = 0;

         }
        }
       }
      }
     } else {
      startTime2 = 0; // Reset the starting time if the limit switch is released
     }
    }
   }
```

```
   else if (v3 < 1) { // OPen Chamber 3

    digitalWrite(relay3Pin, HIGH);
    digitalWrite(RELAY_charging3, LOW); // normaly closed relay on charger is now closed and current flowing to charger
```

```
Serial.println("unit3 opened");
unit3 = 1;
/*if (limitSwitch3State == HIGH) {
  timer3 = 0;
  while (timer3 < 2000 && limitSwitch3State == LOW) {
    limitSwitch3State = digitalRead(limitSwitch3Pin);
    //limitSwitch3State = !limitSwitch3State;
    delay(10);
    timer3 += 10;
  }
  }*/
while (unit3 == 1) {
  limitSwitch3State = digitalRead(limitSwitch3Pin);
  delay(100);
  if (limitSwitch3State == HIGH) { // Limit switch 3 is pressed
    Serial.println("Limit switch 3 is pressed");
    if (startTime3 == 0) {
      startTime3 = millis(); // Record the starting time of the limit switch press
    }

    if (millis() - startTime3 >= 500) { // Check if 2 seconds have elapsed
      if (relay3StartTime == 0) {
        relay3StartTime = millis(); // Record the starting time of relay 3 activation
        volt3 = analogRead(A2);// read the input
        v3 = map(volt3, 0, 1023, 0, 2500) + offset; // map 0-1023 to 0-2500 and add correction offset
        v3 /= 100; // divide by 100 to get the decimal values
```

```
if (v3 > 1) {
  Serial.println("Chamber 3 Closed");
  digitalWrite(relay3Pin, LOW);// Trip relay 3 if voltage is more than 2 volts for voltage sensor 1
  if (v1 >= v2) {


    digitalWrite(RELAY_charging1, HIGH); // normaly closed relay on charger is now open and no current flowing to empty chamber
    digitalWrite(relay1Pin, HIGH);
    Serial.println("Chamber 1 open");
    delay(3000);
    digitalWrite(relay1Pin, LOW);
    Serial.println("Chamber 1 closed");
    relay1StartTime = 0; // Reset the relay 1 starting time
    unitflag1 = 0;
    unitflag3 = 1;
    unit3 = 0;
  }
  else if (v2 > v1) {


    digitalWrite(RELAY_charging2, HIGH); // normaly closed relay on charger is now open and no current flowing to empty chamber
    digitalWrite(relay2Pin, HIGH);
    Serial.println("Chamber 2 open");
    delay(3000);
    digitalWrite(relay2Pin, LOW);
    Serial.println("Chamber 2 closed");
    relay1StartTime = 0; // Reset the relay 1 starting time
```

```
            unitflag2 = 0;

            unitflag3 = 1;

            unit3 = 0;


           }


          }


         }

        }

       } else {

        startTime3 = 0; // Reset the starting time if the limit switch is released

       }


     }

    }

   }



// Temperature Check

  if (t1 > 35) {

    digitalWrite(RELAY_charging1, HIGH); // normaly closed relay on charger is now open and no current flowing to empty chamber

  } else {

    digitalWrite(RELAY_charging1, LOW); // normaly closed relay on charger is now CLOSED and  current flowing to charger

  }
```

```
if (t2 > 35) {
  digitalWrite(RELAY_charging2, HIGH); // normaly closed relay on charger is now open and no current flowing to empty chamber
} else {
  digitalWrite(RELAY_charging2, LOW); // normaly closed relay on charger is now CLOSED and  current flowing to charger
}


if (t3 > 35) {
  digitalWrite(RELAY_charging3, HIGH); // normaly closed relay on charger is now open and no current flowing to empty chamber
} else {
  digitalWrite(RELAY_charging3, LOW); // normaly closed relay on charger is now CLOSED and  current flowing to charger
}


 //Publish data to Arduino Mega's serial port
  mySerial.print("1:");
  mySerial.print(t1);
  mySerial.print("°C");
  mySerial.print(",");
  mySerial.print(h1);
  mySerial.print("%");
  mySerial.print(",");
  mySerial.print((v1/4.17)*100);
  mySerial.print("%");
  mySerial.print(",");
  mySerial.print(digitalRead(RELAY_charging1));
  mySerial.print(";");
```

```
  mySerial.print(t2);
  mySerial.print("°C");
 mySerial.print(",");
  mySerial.print(h2);
  mySerial.print("%");
  mySerial.print(",");
  mySerial.print((v2/4.17)*100);
  mySerial.print("%");
  mySerial.print(",");
  mySerial.print(digitalRead(RELAY_charging2));
  mySerial.print(";");
  mySerial.print(t3);
  mySerial.print("°C");
  mySerial.print(",");
  mySerial.print(h3);
  mySerial.print("%");
  mySerial.print(",");
  mySerial.print((v3/4.17)*100);
  mySerial.print("%");
  mySerial.print(",");
  mySerial.print(digitalRead(RELAY_charging3));
  mySerial.println(";");


/* Serial.print("Sensor 1 Temperature: ");
  Serial.print(t1);
  Serial.print(" *C, Humidity: ");
```

```
    Serial.print(h1);
    Serial.print("%");

    Serial.print(", Sensor 2 Temperature: ");
    Serial.print(t2);
    Serial.print(" *C, Humidity: ");
    Serial.print(h2);
    Serial.print("%");

    Serial.print(", Sensor 3 Temperature: ");
    Serial.print(t3);
    Serial.print(" *C, Humidity: ");
    Serial.print(h3);
    Serial.print("%");
  */
 /* Serial.print("Voltage Sensor 1: ");
  Serial.print((v1/4.17)*100);
  Serial.print("%");
  Serial.print(", Voltage Sensor 2: ");
  Serial.print(v2);
  Serial.print("V");
  Serial.print(", Voltage Sensor 3: ");
  Serial.print(v3);
  Serial.print("V");
  Serial.println();  */
  delay(2000); // Wait for 1 seconds before taking the next reading
}
```

APPENDIX A.2 – NODEMCU CODE

```
#include <ESP8266WiFi.h>
#include <PubSubClient.h>

String data;
const char* ssid = "TEdata24AA4B";
const char* password = "26904950";
const char* mqtt_server = "91.121.93.94";

WiFiClient espClient;
PubSubClient client(espClient);
unsigned long lastMsg = 0;
#define MSG_BUFFER_SIZE  (50)
char msg[MSG_BUFFER_SIZE];
int value = 0;

void setup_wifi() {
  delay(10);
  Serial.println();
  Serial.print("Connecting to ");
  Serial.println(ssid);

  WiFi.mode(WIFI_STA);
  WiFi.begin(ssid, password);

  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
```

```
    Serial.print(".");
  }

  randomSeed(micros());

  Serial.println("");
  Serial.println("WiFi connected");
  Serial.println("IP address: ");
  Serial.println(WiFi.localIP());
}

void callback(char* topic, byte* payload, unsigned int length) {
  Serial.print("Message arrived [");
  Serial.print(topic);
  Serial.print("] ");
  for (int i = 0; i < length; i++) {
    Serial.print((char)payload[i]);
  }
  Serial.println();

  if (strcmp(topic, "chamber/command") == 0) {
    char command = (char)payload[0];

    if (command == '1') {
      Serial.print(1);
        digitalWrite(LED_BUILTIN, LOW);
        delay(500); // Turn the LED on
```

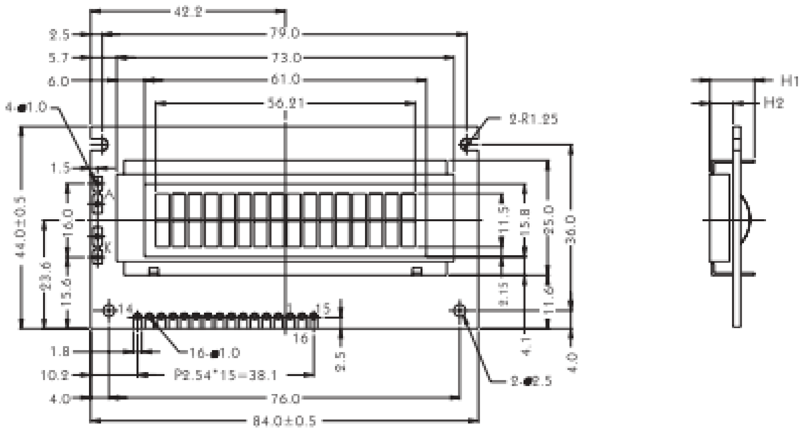
```
      String sensorData = extractSensorData(data, "1:", ';');
      client.publish("chamber/1", sensorData.c_str());
    } else if (command == '2') {
      String sensorData = extractSensorData(data, "2:", ';');
      client.publish("chamber/2", sensorData.c_str());
    } else if (command == '3') {
      String sensorData = extractSensorData(data, "3:", ';');
      client.publish("chamber/3", sensorData.c_str());
    }
  }
}
void reconnect() {
  while (!client.connected()) {
    Serial.print("Attempting MQTT connection...");
    String clientId = "ESP8266Client-";
    clientId += String(random(0xffff), HEX);
    if (client.connect(clientId.c_str())) {
      Serial.println("connected");
      client.publish("sensor/temperatures", "MQTT Server is Connected");
      client.subscribe("chamber/command");
    } else {
      Serial.print("failed, rc=");
      Serial.print(client.state());
      Serial.println(" try again in 5 seconds");
      delay(5000);
    }
  }
```

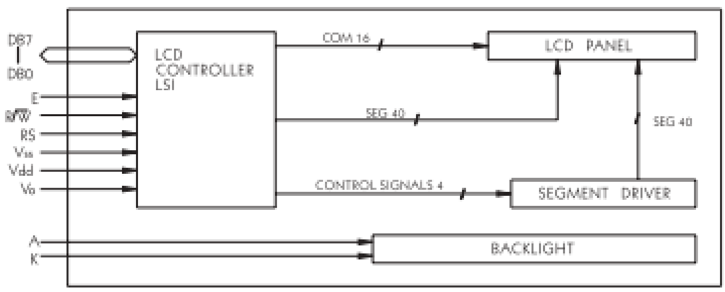
```
}
String extractSensorData(String data, String sensorId, char delimiter) {
  int start = data.indexOf(sensorId);
  if (start != -1) {
    int end = data.indexOf(delimiter, start);
    if (end != -1) {
      String sensorData = data.substring(start, end);
      return sensorData;
    }
  }
  return "";
}
void setup() {
  Serial.begin(9600);
  setup_wifi();
  client.setServer(mqtt_server, 1883);
  client.setCallback(callback);
}
void loop() {
  if (!client.connected()) {
    reconnect();
  }
  client.loop();
  if (Serial.available() > 0) {
    data = Serial.readStringUntil('\n');
    int start = data.indexOf("1:");
    if (start != -1) {
```

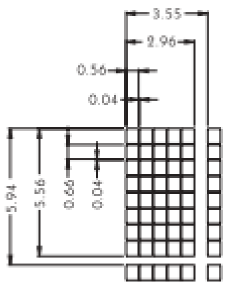
```
      int end = data.indexOf(';', start);
      if (end != -1) {
        String chamberData = data.substring(start, end);
        String sensorData = extractSensorData(chamberData, "1:", ',');
        client.publish("sensor/data", sensorData.c_str());
      }
    }
    sart = data.indexOf("2:");
    if (start != -1) {
      int end = data.indexOf(';', start);
      if (end != -1) {
        String chamberData = data.substring(start, end);
        String sensorData = extractSensorData(chamberData, "2:", ',');
        client.publish("sensor/data", sensorData.c_str());
      }
    }
    start = data.indexOf("3:");
    if (start != -1) {
      int end = data.indexOf(';', start);
      if (end != -1) {
        String chamberData = data.substring(start, end);
        String sensorData = extractSensorData(chamberData, "3:", ',');
        client.publish("sensor/data", sensorData.c_str());
      }
    }
  }
}
```

APPENDIX B – DATASHEETS

APPENDIX B.1 – ARDUINO MEGA 2560







The tolerance unless classified ±0.3mm

MECHANICAL SPECIFICATION			
Overall Size	84.0 * 44.0	Module	H2 / H1
View Area	61.0 * 15.8	W/O B/L	5.1 / 9.7
Dot Size	0.56 * 0.66	EL B/L	5.1 / 9.7
Dot Pitch	0.60 * 0.70	LED B/L	9.4 / 14.0

PIN ASSIGNMENT

Pin no.	Symbol	Function
1	Vss	Power supply (GND)
2	Vdd	Power supply (+5V)
3	Vo	Contrast Adjust
4	RS	Register select signal
5	R/W	Data read /write
6	E	Enable signal
7	DB0	Data bus line
8	DB1	Data bus line
9	DB2	Data bus line
10	DB3	Data bus line
11	DB4	Data bus line
12	DB5	Data bus line
13	DB6	Data bus line
14	DB7	Data bus line
15	A	Power supply for LED B/L (+)
16	K	Power supply for LED B/L (−)

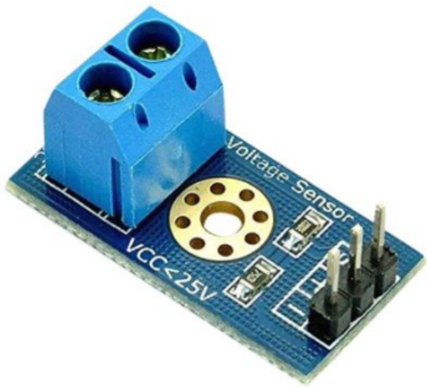
ABSOLUTE MAXIMUM RATING

Item	Symbol	Conditions	Min.	Max.	Unit
Power Supply Voltage	Vdd−Vss	−	0	7	V
LCD Driving Supply Voltage	Vdd−Vee	−	0	13	V
Input Voltage	Vin	−	−0.3	Vdd+0.3	V
Operating Temperature	Topr	Nor.	0	50	°C
Storage Temperature	Tstg	Nor.	−20	+70	°C

ELECTRICAL CHARACTERISTICS (Vdd = +5V, Ta = 25°C)

Item	Symbol	Conditions	Min.	Typ.	Max.	Unit
Logic Supply Voltage	Vdd	−	4.5	5	5.5	V
"H" Input Voltage	V_{IH}	−	2.2	−	−	V
"L" Input Voltage	V_{IL}	−	−	−	0.6	V
"H" Output Voltage	V_{OH}	−	2.4	−	−	V
"L" Output Voltage	V_{OL}	−	−	−	0.4	V
Supply Current	Idd	−	2	−	−	mA
LCD Driving Voltage	V_{LCD}	Vdd−Vo	4.3	−	4.8	V

APPENDIX B.2 – VOLTAGE SENSOR



The Voltage Sensor Module is a simple but very useful module that uses a potential divider to reduce an input voltage by a factor of 5. The 0-25V Voltage Sensor Module allows you to use the analog input of a microcontroller to monitor voltages much higher than it is capable of sensing.

Features & Specifications

1. Output Type: Analog
2. Input Voltage (V): 0 to 25
3. Voltage Detection Range (V): 0.02445 to 25
4. Analog Voltage Resolution (V): 0.00489
5. Dimensions: 4 × 3 × 2 cm

Voltage Sensor Module Pinout

The voltage sensor module has 5 pins, 2 on the front side and 3 on the backside.

1. ***VCC:*** *Positive terminal of the External voltage source (0-25V)*
2. ***GND:*** *Negative terminal of the External voltage source*
3. ***S:*** *Analog pin connected to Analog pin of the microcontroller*
4. ***+:*** *Not Connected*
5. ***-:*** *Ground Pin connected to GND of microcontroller*

APPENDIX B.3 – LIMIT SWITCH

E47 Precision Switches

Product Description

E47 Precision Switches from Eaton's electrical sector provide high accuracy switching at an affordable price. A variety of standard features, such as current capacity, operating force, travel characteristics and actuators, lets you custom fit the switch to your application.

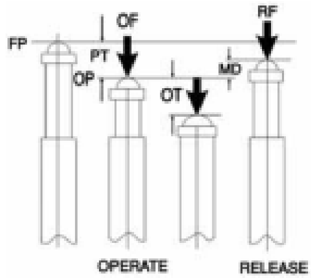
The switches are available in their compact basic form, or enclosed in a rugged, metal housing.

Features

- Compact housings are ideal for use where space is restricted
- Precision, snap-action operators provide accurate repeatability of electrical and mechanical operating characteristics
- High current capacity (up to 20A) allows power load switching and motor handling capability
- Enclosed booted versions shield actuators from debris

Operating Characteristics

Definitions



- OF—Operating Force
- RF—Return Force
- PT—Pre-Travel
- OT—Over-Travel
- MD—Movement Differential
- FP—Free Position
- OP—Operating Position

APPENDIX B.4 – LOCKS

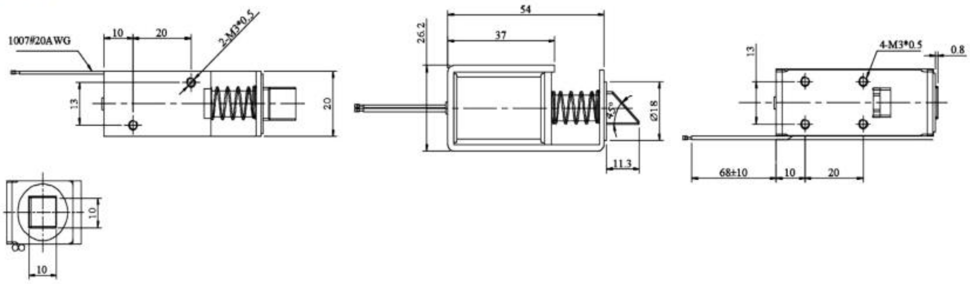
Specifications

Operating Temperature / Humidity	: -20°C to +45°C / 5% to 95% RH
Store Temperature / Humidity	: -20°C to +65°C / 5% to 60% RH
Operating Voltage	: 12V DC ±10%

Electrical Specifications

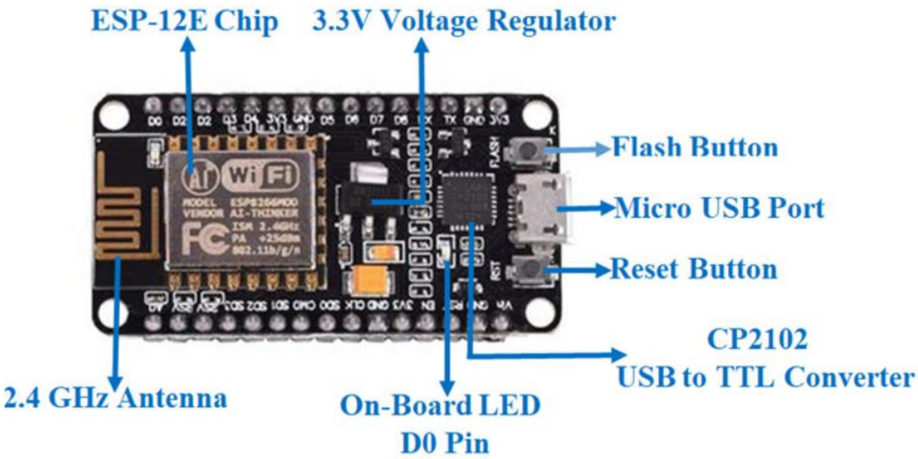
Insulation Resistance	: 500V DC, ≥50MΩ
Dielectric Strength	: 700V AC 50/60Hz
Insulation Level	: Class B (130°C)
Wattage	: 9W (12V DC, R=16Ω ±10%)
Stroke-Force	: 6mm thrust: ≥50gf (12V DC)
Work Cycle	: Pass 0.05 seconds, break 0.05 seconds, max. power-on time, 10 seconds (ED 50%)
Temperature Rise	: ≤80°C (12V DC, 0.05 seconds off for 0.05 seconds, no load)
Response Time	: ≥50mS (12V DC, S=10.5mm, no load)
Leading strength	: 1Kgf-30 seconds
Life	: ≥500,000 times (12V DC, pass for 0.05 seconds, break 0.05 seconds for one time, load (institution)

Diagram



Dimensions : Millimetres

APPENDIX B.5 – VOLTAGE NODEMCU



NodeMCU ESP8266 Specifications & Features

- Microcontroller: Tensilica 32-bit RISC CPU Xtensa LX106
- Operating Voltage: 3.3V
- Input Voltage: 7-12V
- Digital I/O Pins (DIO): 16
- Analog Input Pins (ADC): 1
- UARTs: 1
- SPIs: 1
- I2Cs: 1
- Flash Memory: 4 MB
- SRAM: 64 KB
- Clock Speed: 80 MHz
- USB-TTL based on CP2102 is included onboard, Enabling Plug n Play
- PCB Antenna
- Small Sized module to fit smartly inside your IoT projects

APPENDIX C – SURVEY

ELECTRIC SCOOTERS SURVEY

This survey is for graduation project purposes.

Age *

Short answer text

Gender *

- Male
- Female

Do you drive? *

- Yes, a car
- Yes, a scooter/motorcycle
- No

Do you have an electric vehicle? *

- Electric Car
- Electric Scooter/ motorcycle
- Both
- None

Do you consider buying an electric scooter? *

- ◯ Yes
- ◯ No

Why? *

Long answer text

If a solution was found for the challenges facing electric scooters (such as the small range and the charging time), would you buy an electric scooter? *

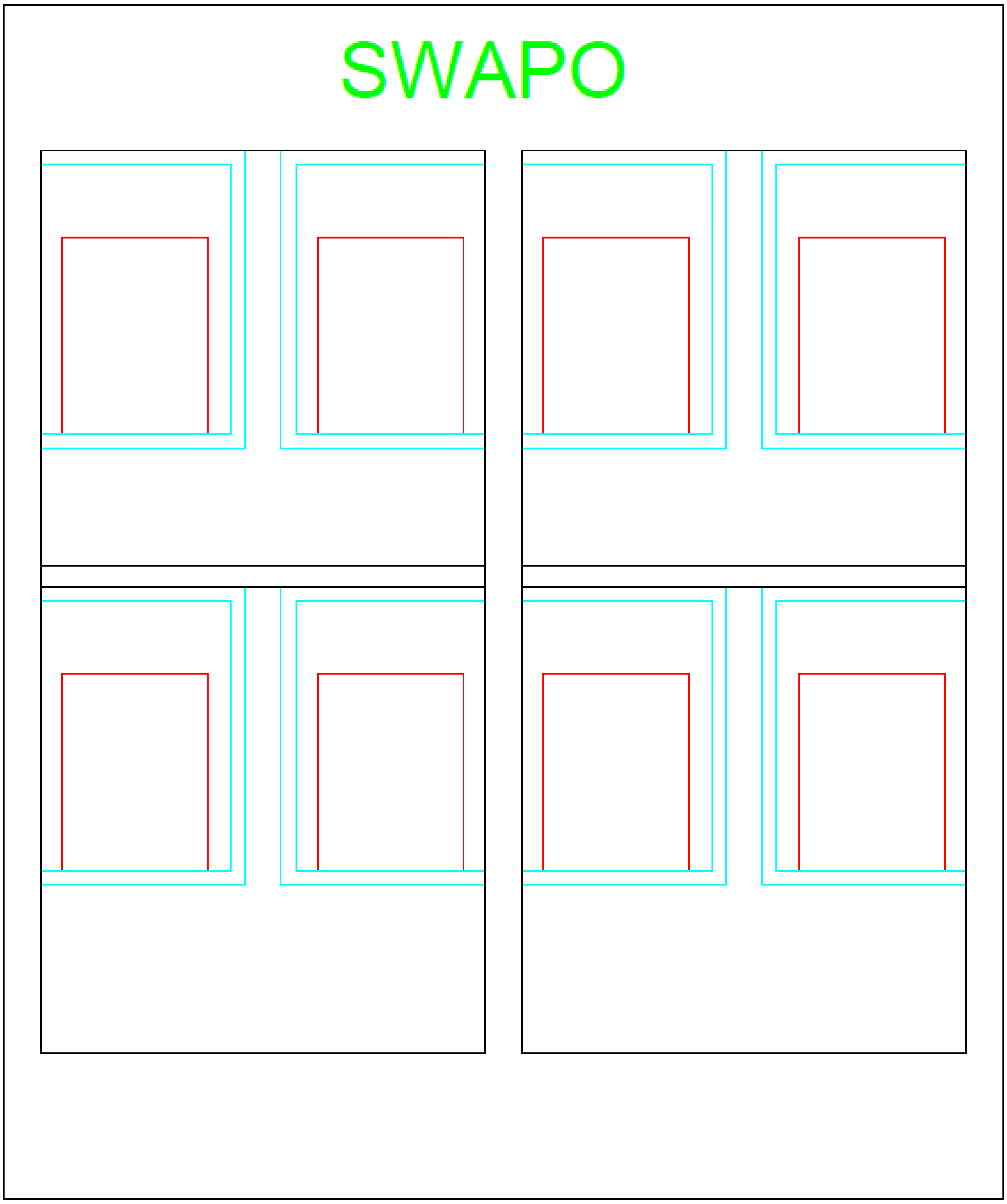
- ◯ Yes
- ◯ No

If no, please state why?

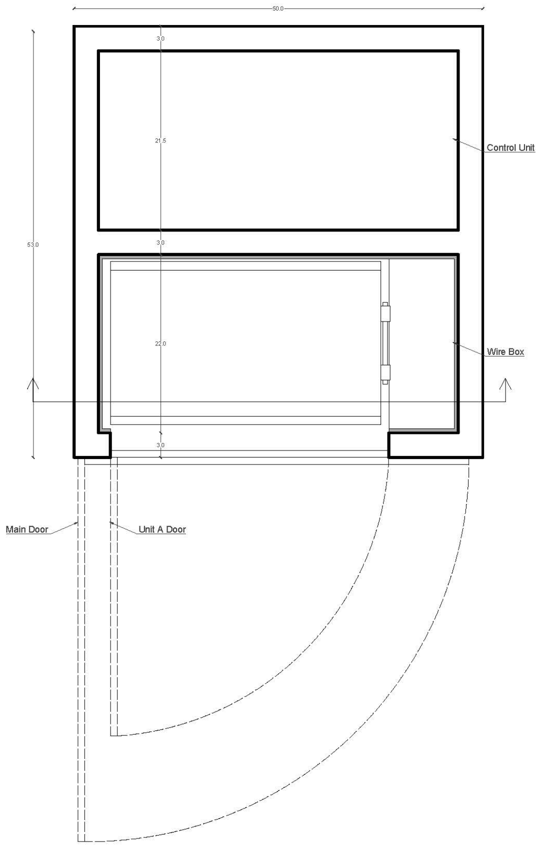
Long answer text

APPENDIX D – DESIGNS

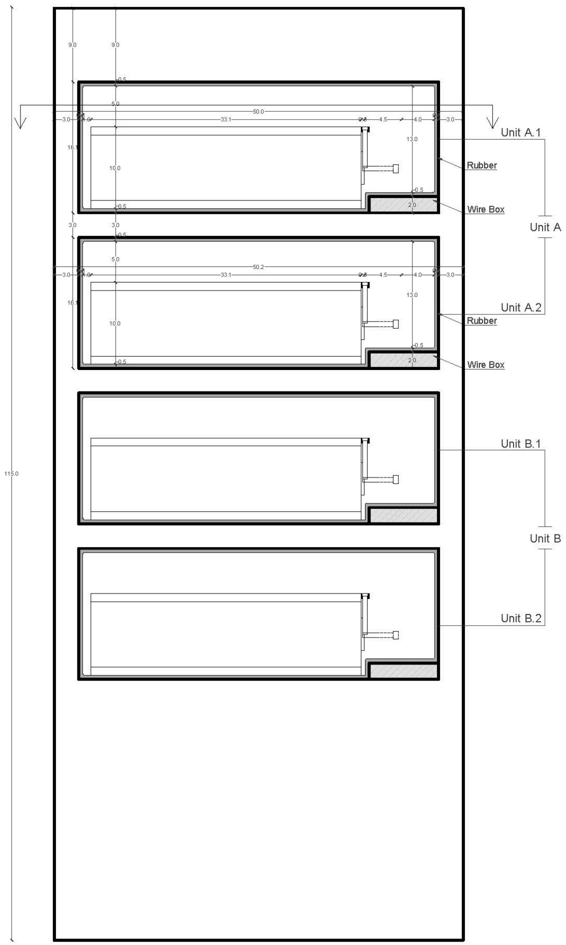
APPENDIX D.1 – DESIGN 1



APPENDIX D.2 – DESIGN 2

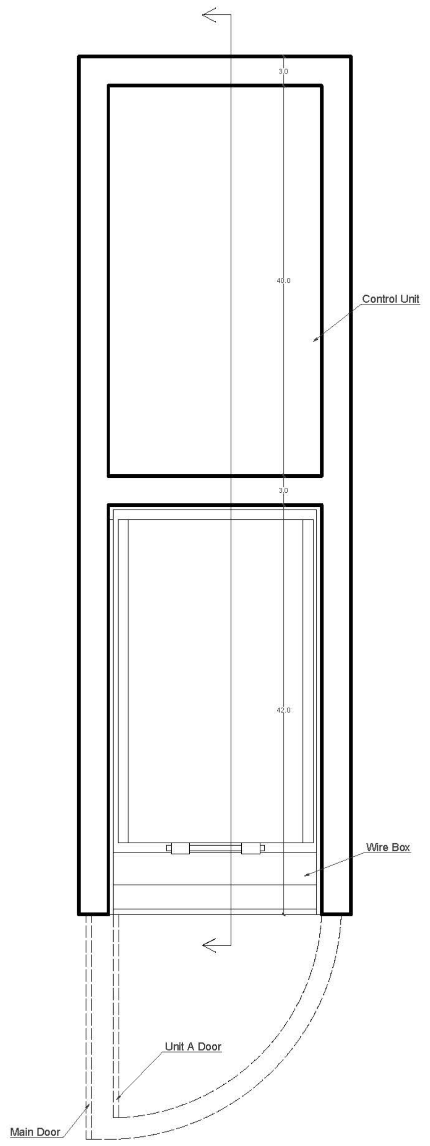


Schematic Plan
Scale 1:5 on A3 paper

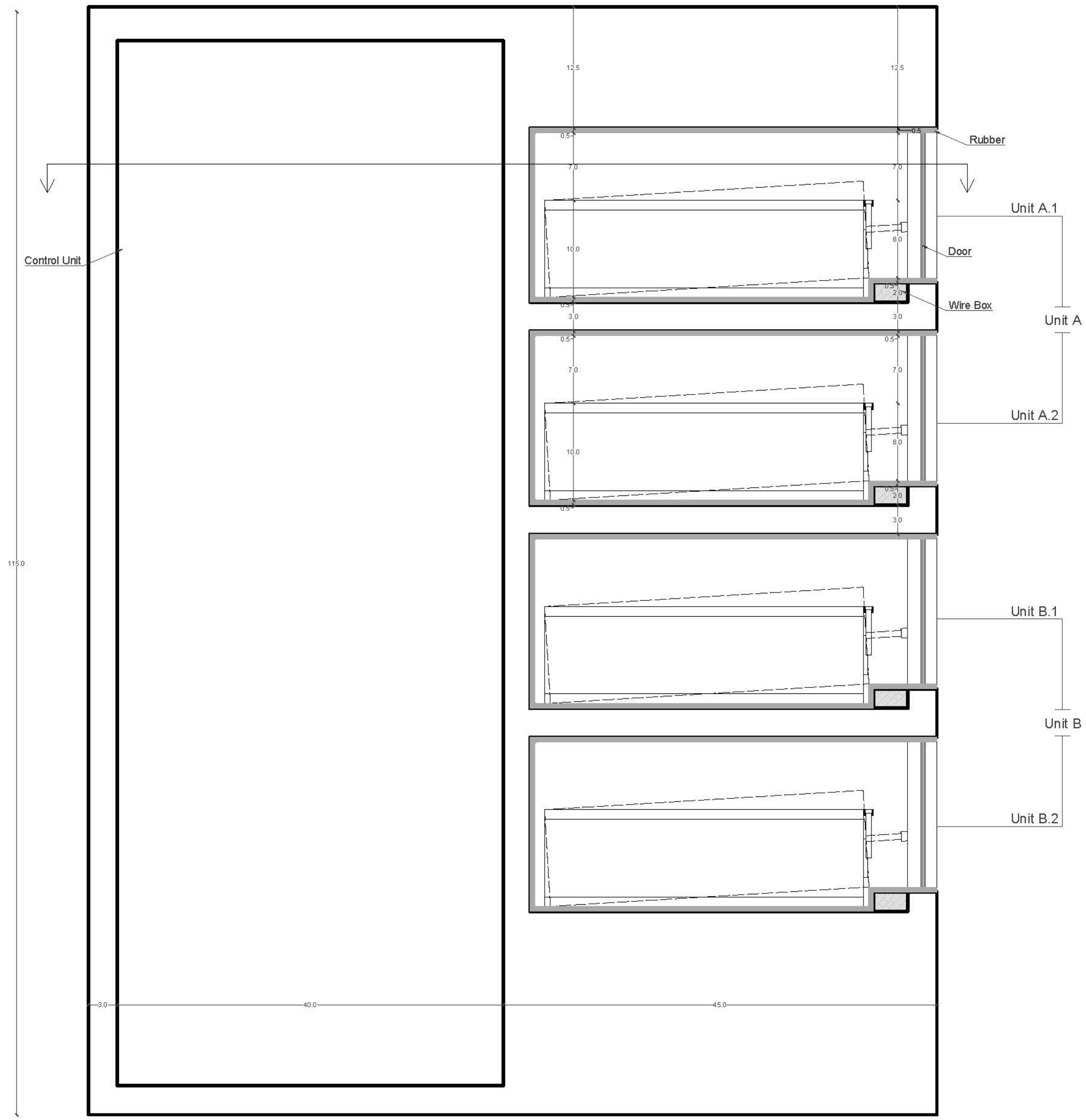


Schematic Section
Scale 1:5 on A3 paper

APPENDIX D.3 – DESIGN 3

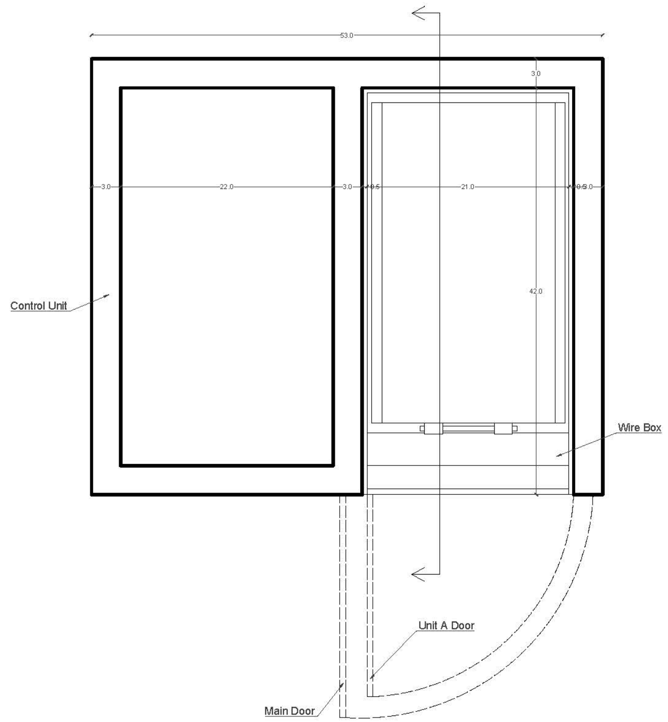


Schematic Plan
Scale 1:5 on A3 paper

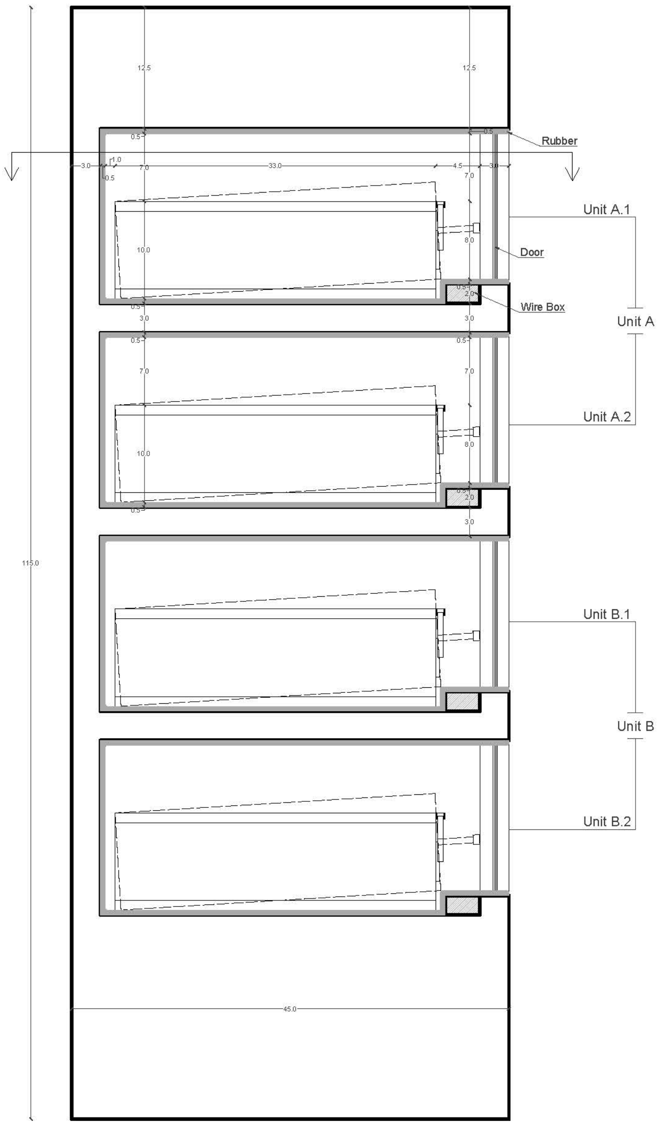


Schematic Section
Scale 1:5 on A3 paper

APPENDIX D.4 – DESIGN 4



Schematic Plan
Scale 1:5 on A3 paper



Schematic Section
Scale 1:5 on A3 paper

APPENDIX D.5 – SCHEMATIC DIAGRAM